



A survey of transformer networks for time series forecasting

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ABSTRACT

Time-series data support critical functions in forecasting, anomaly detection, resource optimization, and real-time decision making across finance, healthcare, energy systems, and networked computing. Classical statistical approaches and early deep-learning architectures (RNNs, LSTMs, CNNs) have achieved notable progress, yet they exhibit structural limitations: recurrent models struggle with long-range dependencies, convolutional models require deep stacking to enlarge receptive fields, and both scale suboptimally to high-dimensional and high-volume data. Transformer architectures—characterized by global attention mechanisms, flexible temporal receptive fields, and growing adoption within foundation-model paradigms—have consequently gained increasing prominence for modern time-series analysis. Drawing on a systematic review of IEEE Xplore, ACM Digital Library, and Scopus (2020–2025), this survey offers a unified, theoretically grounded synthesis of Transformer-based methods for time-series learning. The survey summarizes defining characteristics of time-series data and analyzes core architectural elements, including attention formulations, encoder–decoder structures, hyperparameter design, and domain-specific adaptations. An architecture-centered and task-aware taxonomy is presented to organize recent advances across forecasting, representation learning, anomaly detection, and multimodal fusion. Persistent challenges—spanning computational scalability, data-efficiency constraints, distributional heterogeneity, overfitting risks, hyperparameter instability, interpretability, and reproducibility—are examined in depth. A forward-looking research agenda is outlined, highlighting opportunities in physics-informed architectural design, hybrid neural–mechanistic modeling, resource-efficient real-time inference, multi-resolution spatiotemporal learning, and emerging human–AI collaborative paradigms. By consolidating these methodological developments, this survey aims to provide a structured reference point for ongoing research on Transformer models for time-series machine learning.

Abbreviations

ACC	Accuracy	MASE	Mean absolute scaled error
AI	Artificial intelligence	MCC	Matthews correlation coefficient
AOT	Ahead-of-time	MGT	Meta graph Transformer
AutoML	Automated machine learning	MOF	Metal-organic framework
AUC	Area under the curve	MSE	Mean squared error
AUROC	Area under the receiver operating characteristic Curve	MTS	Multivariate time series

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AUPRC	Area under the precision-recall curve	NAS	Neural architecture search
BERT	Bidirectional encoder representations from transformers	NLP	Natural language processing
Bi-STAT	Bimodal statistical Transformer	NMAE	Normalized mean absolute error
COT	Chain of thought	NRMSE	Normalized root mean squared error
Claim-PT	Claim pre-training	PBT	Population-based training
DAIFT	Domain adaptive intermediate fine-tuning	ReLU	Rectified linear unit

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DAST	Dual-aspect self-attention Transformer	RL	Reinforcement learning
DCNN	Deep convolutional neural network	RMSE	Root mean squared error
DSE	Design space exploration	RMSLE	Root mean squared logarithmic error
ECG	Electrocardiogram detection	RNN	Recurrent neural networks
DETR	Transformer		
EHR	Electronic health record	RUL	Remaining useful life
EVs	Electric vehicles	SMAPE	Symmetric mean absolute percentage error
FLOPS	Floating point operations per second	SOC	State of charge
HI-BEHRT	Hybrid Interpretable biomedical entity-based hierarchical Transformer	SOH	State of health
ILI	Influenza-like illness	SSL	Self-supervised learning
L2M	Least-to-most	TFT	Temporal fusion Transformer
Light-GBM	Light gradient boosting machine	TST	Time series Transformer
LSTM	Long short-term memory	ViT	Vision Transformer
MAE	Mean absolute error	WMAPE	Weighted mean absolute percentage error
MAPE	Mean absolute percentage error	XAI	Explainable artificial intelligence

1. Introduction

Time series analysis and forecasting [1] are essential tools across a wide range of fields, including economics, healthcare, energy, and transportation. These methods focus on analyzing data collected at regular intervals to identify meaningful patterns and trends over time [2]. Such insights are critical for anticipating future outcomes and informing decision-making. Traditionally, time series forecasting has relied on statistical models such as autoregressive integrated moving average [3,4] and exponential smoothing [5]. While classical models are well-suited for capturing recurring patterns and gradual shifts in time series data, they often fall short when applied to sequences characterized by nonlinearity, noise, or extended temporal dependencies. To address these challenges, machine learning techniques have been introduced as more flexible alternatives, offering the ability to learn intricate relationships without relying on rigid assumptions [6]. Recently, deep learning models have demonstrated strong performance in sequence modeling tasks [7]. Among them, recurrent neural networks (RNNs) and long short-term memory (LSTM) networks are widely adopted for their ability to capture dynamic temporal dependencies [8].

The introduction of Transformers, as detailed in the seminal 2017 paper by Google [9], “*Attention Is All You Need*,” marked a turning point in machine learning. Initially designed for language translation, this architecture has demonstrated remarkable versatility, extending its influence across a wide range of AI domains. In parallel, time series analysis has long been a core tool in fields such as finance, energy systems, healthcare, climate science, and industrial monitoring, where the goal is to model how variables evolve over time for forecasting, anomaly detection, and decision support. Traditional approaches—including statistical models such as ARIMA and state-space methods, as well as early deep learning models based on RNNs and LSTMs—have achieved important successes but often struggle with long-range dependencies, irregular patterns, and the growing scale of modern datasets. Transformers fundamentally change the way time series can be modeled by offering improved prediction accuracy and more flexible handling of complex sequential data than these earlier methods. Unlike RNNs and LSTMs, which process data step by step, Transformers use an attention mechanism that considers all time steps in parallel [10], enabling the

model to focus on the most relevant points in the history regardless of their position [11]. This ability to capture long-range dependencies and to scale effectively with large datasets makes Transformers especially suitable for challenging time series forecasting tasks, such as multi-horizon demand prediction or long-term energy load planning. As the field of time series analysis advances, the adoption of these advanced models is not only improving forecast accuracy but also expanding analytical capabilities to reveal deeper and more subtle temporal patterns that were difficult to detect with traditional techniques. This shift is reshaping data-driven decision-making across many sectors, underscoring the importance of continued innovation in time series modeling and forecasting.

Existing surveys have established a strong foundation for deep learning-based time series forecasting, offering comprehensive benchmarks of core architectures across diverse data types and prediction horizons [12]. Long sequence forecasting has been thoroughly examined, with clear taxonomies and analyses addressing dependency modeling and computational efficiency [13]. Recent efforts have introduced prompt-based paradigms, reformulating forecasting as a language generation task with demonstrated generalization under zero-shot conditions [14]. Comparative studies on channel-independent and channel-dependent strategies have revealed critical robustness–capacity trade-offs, providing actionable insights into model stability under distributional shifts [15]. Transformer-specific reviews have synthesized advances in input encoding, attention design, and architectural tailoring for temporal data [16].

While these contributions are significant, they primarily focus on model-level comparisons and lack a unified, task-oriented framework linking Transformer variants to specific forecasting scenarios. To address this gap, this survey presents a systematic, application-driven analysis of Transformer-based methods, highlighting domain-specific adaptations, categorizing task-level requirements, and evaluating recent innovations in efficiency, interpretability, and transferability. To ensure comprehensive and balanced coverage, this survey integrates (1) a structured search of peer-reviewed studies published between 2020 and 2025 across IEEE Xplore, ACM Digital Library, and Scopus; (2) influential preprints and arXiv submissions that reflect the latest advancements not yet formally published; and (3) seminal works prior to 2020 that established foundational paradigms in deep learning, sequence modeling, and time series forecasting. Studies were filtered and categorized by model architecture, domain application, and forecasting objective. To visualize the state-of-the-art developments and interconnections within the field, we present a bibliometric network of Transformer-based time series forecasting (Fig. 1), offering a comprehensive overview of the current research. Furthermore, this survey introduces a task-driven taxonomy and identifies open research questions related to generalization, scalability, and the integration of foundation models. By synthesizing methodological advances and pinpointing gaps in current practice, this work aims to inform both future research and real-world deployment.

This survey reviews Transformer-based methods for time series forecasting and aims to advance both theory and practice, as outlined in Fig. 2 and reflected in the structure of the paper. Section 1 introduces the topic, explains why this study is needed, and summarizes the main contributions. Section 2 presents the basic concepts: it first describes the key features of time series data, then introduces the original Transformer architecture for readers who are new to it, and finally examines recent design improvements, including choices of important training settings and adaptations to different application domains. Section 3 then turns to applications and reviews case studies in finance, healthcare, energy systems, and transportation, showing how Transformer models handle long-range temporal patterns and high-dimensional inputs in real-world tasks. Section 4 focuses on open technical issues, including high computational cost, large data requirements, the risk of overfitting (where models fit training data too closely and perform poorly on new data), and the difficulty of tuning many design parameters. Section 5

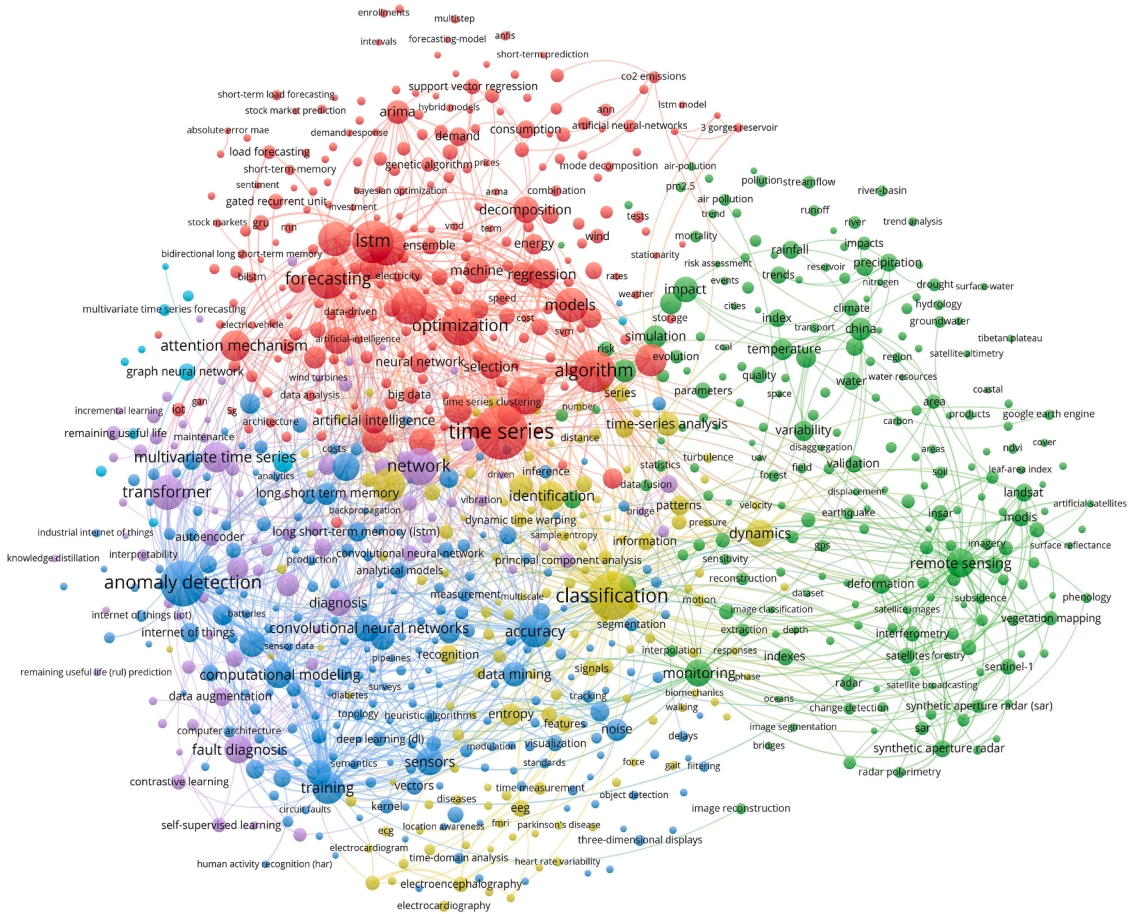


Fig. 1. Bibliometric network visualization of time series analysis and forecasting.

looks ahead and outlines future research directions, such as the use of domain knowledge, the integration of multiple types of data, and automated methods for model design and training. Section 6 concludes the survey by summarizing the main findings and discussing their broader implications. Overall, this survey aims to build a clearer understanding of Transformer-based time series methods and to highlight promising directions for further progress across different disciplines.

2. Theoretical background

Time series data, characterized by temporal dependence, trends, seasonality, and noise, underpins many scientific and industrial applications and requires dedicated methods for reliable forecasting. Conventional models often have difficulty capturing long-range dependencies and scaling to large datasets, whereas Transformer architectures with self-attention offer clear benefits for sequence modeling [17]. Effective use of Transformers in this setting requires addressing non-stationary behavior, long input horizons, and computational constraints, for example through modified self-attention, task-specific positional encodings, more efficient attention variants, and the inclusion of external signals such as economic indicators. Recent architectures such as Informer, Time Series Transformer (TST) [18], and Temporal Fusion Transformer (TFT) [19] build on these ideas and provide specialized designs for forecasting, offering a practical basis for accurate and scalable Transformer-based time series models.

2.1. Characteristics of time series data

Time series data, a sequence of data points collected at successive, equally spaced points in time, is a cornerstone of various scientific and

industrial fields [20,21]. Its unique characteristics necessitate specialized analytical techniques, models, and mining benchmarks [22]. Understanding these characteristics is crucial for effective analysis and forecasting. In engineering applications, these properties are particularly pronounced in safety-critical systems such as solid-rocket thrust profiles, rotating machinery vibration signals, and second-life battery degradation trajectories, where reliable modeling of multi-scale, nonlinear temporal dynamics is essential [23,24]. One fundamental characteristic of time series data is temporal dependence. Unlike cross-sectional data, time series data points are not independent; each observation is influenced by its predecessors [25]. Properly accounting for this dependency structure is critical for accurately capturing the underlying processes generating the data. Ignoring temporal dependence can lead to misleading conclusions and poor predictive performance, as the intrinsic sequential nature of the data plays a vital role in shaping its behavior. Trend is another key feature, representing the long-term movement in the data over time. Trends can be upward, downward, or constant and are essential for understanding the overall direction of the series. Identifying and modeling trends is crucial for making long-term forecasts, as trends often reflect underlying factors such as economic growth, technological advancements, or demographic changes [26]. Accurate trend analysis helps in strategic planning and long-term decision-making.

Seasonality refers to regular, repeating patterns or cycles within the data at fixed intervals, such as daily, monthly, or yearly [27]. Seasonal effects are common in various applications, such as retail sales, where demand fluctuates during holidays, or electricity consumption, which varies with weather patterns. Accurately identifying and accounting for seasonality can significantly enhance forecast accuracy by isolating these predictable variations from other components of the time series,

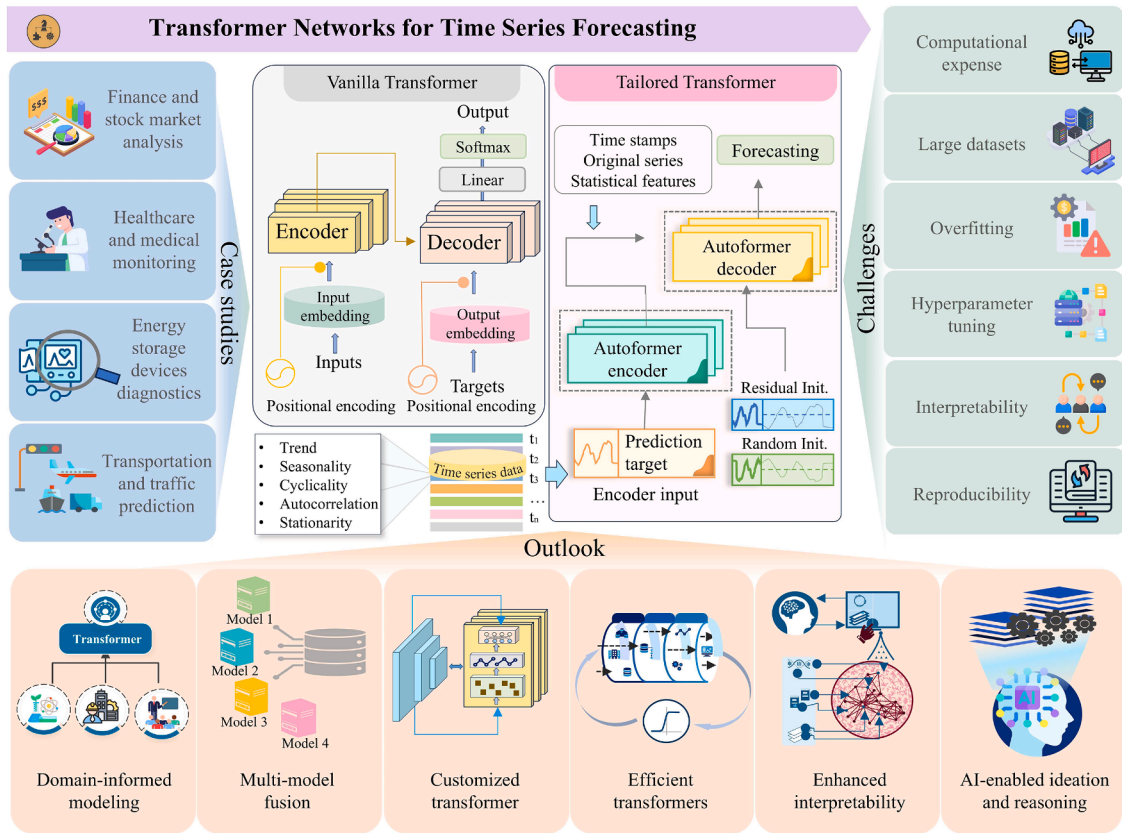


Fig. 2. Architecture of this survey on Transformer models for time-series analysis and forecasting.

allowing for more precise predictions and resource allocation. Cyclic behavior, similar to seasonality but not as regular, involves fluctuations that occur over longer, variable periods [28]. Economic cycles, for example, do not follow a fixed pattern but show alternating phases of growth and contraction. Understanding these cycles can provide insights into the broader temporal dynamics of the data. In contrast to seasonal effects, cycles are shaped by a broader range of external influences—such as political events, market dynamics, and technological innovations—rendering their analysis more intricate. Nevertheless, capturing these cyclical patterns remains essential for achieving a comprehensive and accurate interpretation of time series data. Stationarity is a property indicating that the statistical characteristics of the time series, such as mean and variance, do not change over time. Many time series models assume stationarity; thus, transforming non-stationary data into a stationary form (e.g., through differencing) is often a necessary preprocessing step. Stationary data simplifies the modeling process and enhances the robustness of statistical tests and forecasts, providing a stable framework for identifying genuine signals within the data.

Noise or random variation is an inherent part of time series data, representing the unpredictable, random component [29]. Effective models need to distinguish between meaningful patterns and noise to avoid overfitting. Overfitting occurs when a model captures the noise instead of the underlying signal, leading to poor generalization on new data. Techniques such as smoothing and filtering can help reduce the impact of noise, thereby improving the reliability of the forecasts. Autocorrelation is the correlation of a time series with its past values. This self-similarity across time intervals can reveal important temporal patterns and is often visualized using autocorrelation plots. High autocorrelation indicates that past values have a strong influence on future values, which is critical information for building predictive models. Autocorrelation can help identify lagged effects and inform the selection of appropriate model parameters. Finally, addressing missing values and

outliers is crucial in time series analysis. Missing data can arise from various sources, such as equipment failures or recording errors, while outliers may represent unusual events or errors in the data. Properly managing these issues is essential to preserve the accuracy and integrity of the analysis. Approaches such as data interpolation, imputation, and the application of robust statistical methods can effectively mitigate the impact of these challenges, ensuring that the models developed are both reliable and accurate. In summary, time series data is characterized by temporal dependence, trends, seasonality, cyclic behavior, stationarity, noise, autocorrelation, and issues related to missing values and outliers. Recognizing and addressing these characteristics through appropriate modeling techniques is essential for accurate time series analysis and forecasting. Each of these features contributes uniquely to the complexity and richness of time series data, necessitating a comprehensive and nuanced approach to fully harness its predictive potential.

2.2. Overview of Transformer architecture

The Transformer architecture, introduced by Vaswani et al. [9], replaces recurrent structures with self-attention in sequence modeling, allowing parallel processing of input sequences rather than stepwise updates and thereby increasing computational efficiency and shortening training time while maintaining strong modeling capacity. Self-attention is the central mechanism: it dynamically evaluates relationships among elements in an input sequence without relying on their exact positions. By computing attention scores that indicate the relative importance of each element and using these scores to form weighted combinations of features, the model can focus on task-relevant information and capture complex long-range dependencies that conventional models find difficult to learn. Fig. 3 presents a generic Transformer framework and highlights how multi-head attention layers interact with simple feed-forward layers to support this expressive behavior. In the standard configuration, the Transformer consists of an encoder and a decoder,

where the encoder maps an input sequence to a series of continuous representations and the decoder uses these representations to generate the output sequence step by step. Both components are implemented as stacks of identical layers, each containing a multi-head self-attention layer and a feed-forward layer, with shortcut connections and normalization around each layer to stabilize training and improve convergence. Positional encodings compensate for the absence of recurrence by adding information about relative or absolute positions of tokens so that the model can interpret sequence order within a broader temporal or structural context. Taken together, the parallel architecture and the ability to model long-range dependencies have led to wide adoption of the Transformer in many application areas; surveys in natural language processing are provided in [30], adaptations for vision tasks are discussed in [31,32], work on video analysis is reviewed in [33], and an extended summary of attention mechanisms, positional encodings and architectural variants is given in [11].

2.2.1. Encoder

The encoder section in a transformer model plays a crucial role in processing input sequences. It consists of several stages, including self-attention, feed-forward neural networks, normalization, and positional encoding, which together help in capturing both local and global dependencies in the data.

(a) Self-attention

In the self-attention mechanism, the input matrix, denoted as X , is transformed into three key components: queries, keys, and values. These components are computed through matrix multiplications, where the weights W_Q , W_K , and W_V are learned during training. The transformation can be expressed as follows:

$$Q = XW^Q, K = XW^K, V = XW^V \quad (1)$$

where Q , K , and V represent the query, key, and value matrices,

respectively. These matrices are crucial for determining the output of the attention mechanism. Next, the attention scores are computed by taking the dot product of the query and key matrices, followed by a softmax operation to normalize the scores. The attention scores A are calculated as:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (2)$$

where d_k is the dimension of the key vector, which normalizes the dot product to control the scale of the attention scores. Finally, the output of the attention mechanism is obtained by multiplying the attention scores with the value matrix V , as shown in the following equation:

$$Z = AV \quad (3)$$

This output Z is a weighted sum of the values, with the attention scores determining the importance of each value.

(a) Feed-forward neural network

Once the attention output is obtained, each position in the sequence is passed through a feed-forward neural network. This network consists of two layers with an activation function (ReLU) in between. The feed-forward process can be mathematically expressed as:

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (4)$$

Here, W_1 and W_2 represent the weight matrices, while b_1 and b_2 correspond to the bias terms for the two layers. This process enables the model to learn non-linear transformations and capture more intricate patterns within the data.

(a) Add & Norm

After the feed-forward network, layer normalization and residual connections are applied to stabilize the training process and prevent

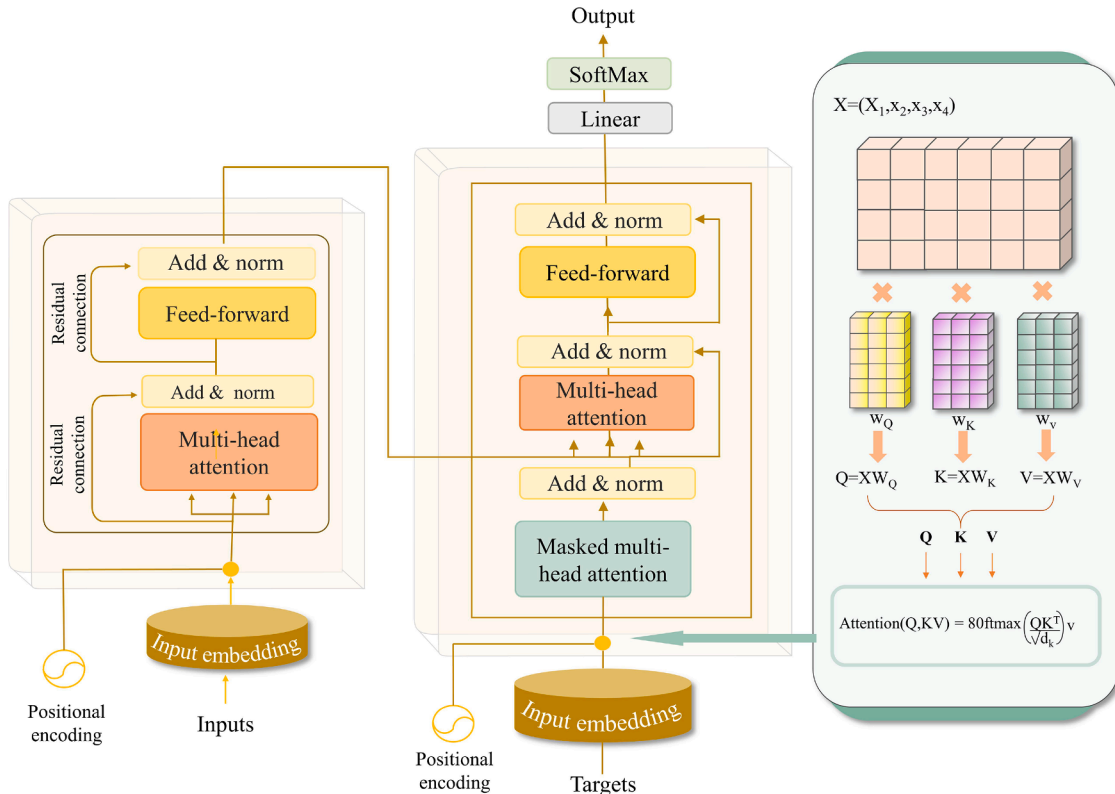


Fig. 3. Overview of a generic Transformer network framework.

vanishing gradients. The residual connection helps preserve the original input information and adds it to the output of the self-attention or feed-forward network, improving convergence. The residual connection and normalization are expressed as:

$$Y = \text{LayerNorm}(X + \text{Sublayer}(X)) \quad (5)$$

where the sublayer refers to either the self-attention or the feed-forward network, depending on the layer.

(a) Positional Encoding

Transformers lack an inherent sense of order in the input sequence, unlike recurrent neural networks. To address this, positional encodings are added to the input representations to provide information about the positions of elements in the sequence. The positional encoding formulas are defined as:

$$PE(pos, 2i) = \sin(pos / 10000^{2i/d_{\text{model}}}) \quad (6)$$

$$PE(pos, 2i + 1) = \cos(pos / 10000^{2i/d_{\text{model}}}) \quad (7)$$

where pos is the position in the sequence, i is the dimension, and d_{model} is the total dimension of the model. These encodings ensure that each position has a unique representation that can be learned by the transformer.

2.2.2. Decoder

The decoder component of the Transformer model is tasked with generating the output sequence by leveraging the encoded representations of the input. While it shares a similar architectural structure with the encoder, the decoder incorporates several critical modifications to support autoregressive sequence generation. Notably, it includes a masked multi-head self-attention mechanism that prevents access to future positions in the sequence, thereby ensuring causal dependency. In addition, an encoder-decoder attention layer is employed to allow the decoder to attend to relevant segments of the encoded input, enabling effective information integration across input and output domains.

(a) Masked self-attention

The self-attention mechanism in the decoder follows the same structure as that in the encoder, with one key difference: a causal mask is applied to block access to future positions. This ensures that each prediction depends only on earlier tokens, thereby preserving the autoregressive nature of the model. The masked attention is formally defined as:

$$A' = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right) \quad (8)$$

where M represents a mask that applies a penalty of $-\infty$ to future positions, making their softmax values equal to zero. This prevents the model from attending to future positions during training.

(a) Encoder-decoder attention

The encoder-decoder attention mechanism is a crucial step in the decoder, where the queries originate from the previous hidden states of the decoder, and the keys and values are derived from the output of the encoder. This attention mechanism allows the decoder to focus on relevant parts of the input sequence while generating the output sequence. The encoder-decoder attention is computed as:

$$A'' = \text{softmax}\left(\frac{Q_{\text{dec}}K_{\text{enc}}^T}{\sqrt{d_k}}\right) \quad (9)$$

The output of this attention is then obtained by multiplying the

attention scores with the values derived from the encoder output:

$$Z_{\text{dec}} = A''V_{\text{enc}} \quad (10)$$

This allows the decoder to selectively attend to different parts of the encoded input sequence while generating each token in the output sequence.

(a) Feed-forward and add & Norm in decoder

Following the attention layers, the decoder applies a position-wise feed-forward neural network and layer normalization, mirroring the structure used in the encoder. These components operate independently at each position and are essential for introducing non-linearity and stabilizing training. In the decoder, they refine representations by integrating contextual signals from both the encoder output and previously generated tokens.

2.3. Hyperparameters optimization of Transformers

Hyperparameter tuning and optimization constitute critical processes in enhancing the performance of Transformer models across a wide range of tasks [34]. These hyperparameters—encompassing architectural configurations such as the number of layers, attention heads, and embedding dimensions, as well as training parameters like learning rate, batch size, and dropout rate—directly influence the capacity of the model to learn meaningful representations. Careful calibration of these parameters is essential for ensuring convergence stability, generalization ability, and computational efficiency, thereby shaping the overall effectiveness of the model. Adjusting them carefully can lead to substantial improvements in both accuracy and operational efficiency. For instance, the number of layers in a Transformer is a key determinant of its information processing capacity [35]. Each layer facilitates an additional round of self-attention and feed-forward operations, enabling the model to capture increasingly complex patterns within the data. However, while expanding the number of layers enhances the ability to understand intricate dependencies, it also heightens the risk of overfitting, particularly when data is scarce. Moreover, adding layers increases computational demands, which can negatively affect both training speed and resource consumption. Therefore, achieving an optimal balance in the number of layers is crucial for developing a model that maintains both high accuracy and efficiency.

The configuration of attention heads [36] and the dimensionality of the feed-forward network [37] are critical hyperparameters in Transformer models, directly affecting the capacity of the model to analyze and interpret various segments of the input data concurrently. Increasing the number of attention heads enhances the ability to simultaneously process multiple perspectives of the input data, enriching its comprehension and representation capabilities. This multiplicity allows Transformers to discern and integrate diverse contextual relationships within the data, which is particularly valuable in complex tasks involving nuanced interpretations. Conversely, the dimensionality of the feed-forward network defines the computational breadth of each layer, influencing how robustly the model can process the information. Higher dimensionality equates to a greater capacity for handling information, allowing the network to learn more detailed and subtle features from the input data. However, elevating either the number of attention heads or the network dimensionality increases the complexity and computational requirements of the learning machine. This escalation can lead to models that are computationally expensive and require extensive datasets for effective training to avoid overfitting. Therefore, careful tuning of these parameters is essential to balance model depth and complexity with performance and efficiency, ensuring that the Transformer remains both powerful and practicable across varied applications.

Learning rate is critical in determining how quickly a model adjusts

Table 1

Key Transformer hyperparameters, their functional roles, and practical tuning considerations.

Hyperparameter	Description	Typical range	Practical considerations	References
Number of layers	Number of stacked encoder or decoder blocks. Depth increases model capacity.	2–12 (small); up to 24+ (large-scale)	Deeper models may overfit on limited data and increase training time.	[9,41]
Number of attention heads	Parallel attention units that capture distinct subspaces.	8–16	Increasing heads improves representation capacity but scales memory quadratically.	[9,42]
Embedding dimension	Size of token and positional embeddings.	512–1024	Should match input complexity; divisible by number of heads.	[42]
Feed-forward dimension	Width of the two-layer MLP in each block.	2048–4096	Larger values enhance expressiveness but increase latency. Often set to $4 \times$ embedding dimension.	[42]
Learning rate	Step size for updating weights during training.	$1e-4$ – $1e-3$	Requires careful scheduling (e.g., cosine decay, warm-up) to avoid instability.	[9,42]
Batch size	Number of training samples processed in one update.	16–64 (moderate); 128–512 (large-scale)	Larger batches improve stability but demand more memory.	[42,43]
Dropout rate	Fraction of units randomly deactivated to prevent overfitting.	0.1 – 0.3	Higher dropout helps regularization but may impair learning in small models.	[42,44]
Positional encoding	Injects order information via sinusoidal or learned embeddings.	Sinusoidal (default); learned (advanced)	Learned embeddings can adapt to data structure but require more training.	[46]
Warm-up steps	Initial steps with gradually increasing learning rate.	1000–4000	Stabilizes training in early epochs, especially for large models.	[45]
Attention dropout	Dropout applied to attention weights.	0.1 – 0.2	Helps reduce overfitting in attention maps; often used with standard dropout.	[44]
Optimizer	Algorithm used for weight updates (e.g., Adam, AdamW).	Adam (default); AdamW (improved variant)	Choice affects convergence speed and final performance. Weight decay often tuned with optimizer.	[47]

its weights during training [38]. An appropriately set learning rate helps the model converge to a good solution efficiently, balancing speed and accuracy without overshooting optimal weights. Learning rate schedulers, which adjust the rate according to training progress, play a pivotal role here. For example, “warm-up” strategies gradually increase the learning rate at the beginning of training to prevent the model from settling too quickly into a poor local minimum [39]. Conversely, “cool-down” strategies decrease the rate towards the end of training, allowing the model to fine-tune adjustments to its weights, enhancing model stability and performance [40].

Careful tuning of hyperparameters is critical for adapting Transformer models to specific tasks and datasets. Appropriate configuration enhances generalization to unseen data while improving computational efficiency, thereby enabling more scalable and effective training. Hyperparameter optimization plays a strategic role in determining model performance and is a key factor in the successful deployment of Transformers in real-world applications. The principal hyperparameters are summarized in Table 1.

Hyperparameter optimization is a crucial aspect of deploying Transformer models effectively, as the selection and tuning of these parameters significantly influence model performance. This section explores the intricacies of selecting optimal hyperparameters for Transformer architectures, emphasizing the systematic approach required to balance model complexity with practical constraints such as computational resources and data availability. The process begins with identifying critical hyperparameters that require optimization. Optimizing these hyperparameters involves a delicate trade-off between enhancing the ability of the model to generalize to new data and improving its performance on known data. Advanced strategies for exploring the hyperparameter space include grid search, random search, Bayesian optimization, and more recently, gradient-based optimization techniques.

(a) Grid search and random search:

These traditional methods are foundational to hyperparameter optimization. Grid search methodically evaluates a predefined range of hyperparameter values, systematically assessing each combination to determine the most effective [48]. In contrast, random search selects hyperparameter values randomly within specified bounds, providing a less structured approach that can sometimes yield better results in less time [49]. However, both methods may become impractical and

computationally intensive as the dimensionality of the hyperparameter space increases.

(a) Bayesian optimization

This technique is particularly valuable for high-dimensional parameter spaces, where traditional methods falter. Bayesian optimization builds a probabilistic model that predicts the performance of different hyperparameter settings [50]. This model is then used to intelligently select new hyperparameter combinations to evaluate, focusing on those most likely to offer improvements. This approach not only reduces the number of required evaluations but also tends to find better parameters faster than grid or random search.

(a) Gradient-based optimization

Recently, methods such as hyper-gradient descent [51] and automatic differentiation [52] of hyperparameters have gained popularity. These approaches leverage gradient information to optimize hyperparameters, thus potentially accelerating the optimization process significantly. By adjusting hyperparameters in a direction that maximally improves the performance of the model, these methods can achieve rapid convergence compared to non-gradient techniques.

(a) Evolutionary algorithms and genetic algorithms

Inspired by biological evolution, these algorithms apply mechanisms such as mutation, crossover, and selection to evolve hyperparameters over generations. Evolutionary and genetic algorithms are especially useful for optimizing complex, non-differentiable functions often encountered in model training [53]. These algorithms explore the hyperparameter space extensively and are robust to local optima, making them effective for complex model configurations.

(a) Population-based training

Population-based training (PBT) enhances the efficiency of hyperparameter optimization by leveraging a community of model instances, which evolve together [54]. This method allows dynamic tuning and adaptation of hyperparameters during the training process itself, rather than after predefined epochs. PBT continuously evaluates the performance of each model within the population, promoting

hyperparameters from better-performing models to those that are underperforming. This technique is particularly beneficial for lengthy training processes, where it can significantly reduce the time and resources required by enabling faster convergence on optimal model configurations.

(a) Automated machine learning

Automated machine learning (AutoML) represents a notable advancement in hyperparameter optimization by significantly reducing the reliance on manual configuration [55]. Through the automation of hyperparameter selection, model architecture search, and training strategy adjustment, AutoML frameworks facilitate the efficient development of high-performing models. This not only accelerates the modeling process but also broadens access to advanced machine learning techniques, enabling practitioners to deploy sophisticated solutions with minimal domain-specific intervention. These systems employ sophisticated algorithms to systematically explore various combinations of hyperparameters, meticulously adjusting them based on performance metrics gathered during evaluations against a validation set. This automation not only speeds up the optimization process but also enhances the reproducibility and reliability of the results.

(a) Early stopping and pruning

These techniques are invaluable for managing the complexity and computational demands of large Transformer models. Early Stopping monitors performance metrics such as validation loss during training and terminates the process if the model ceases to improve, thereby saving computational resources and preventing overfitting [56]. Pruning goes a step further by systematically removing less important connections within the neural network, which reduces its complexity and expedites computation without significantly compromising performance [57]. To counteract potential drawbacks of premature termination, incorporating a warm-up period at the beginning of training allows models to stabilize before any critical evaluations or adjustments are made.

(a) Transfer learning

This approach to hyperparameter optimization utilizes optimal hyperparameters identified in one task to inform the tuning process for a related task [58]. By leveraging knowledge gained from previous applications, transfer learning can significantly accelerate the optimization of Transformer models for new tasks, especially when these tasks share underlying similarities. This method is particularly effective when fine-tuning models, as it allows practitioners to bypass some of the more time-consuming phases of the optimization process, assuming the foundational task is sufficiently analogous to the target task.

(a) Meta-learning

Meta-learning, often described as “learning to learn,” entails the development of models capable of rapidly adapting to new tasks with minimal additional data or training effort [59]. Within the domain of hyperparameter optimization, meta-learning shifts the focus from task-specific tuning to the identification of transferable patterns and strategies. By analyzing the performance of hyperparameter configurations across diverse tasks and datasets, meta-learning enables the formulation of generalized principles that guide more efficient and effective optimization processes in future applications. This technique empowers models to predict optimal hyperparameters for any given task, based on its characteristics and the accumulated knowledge from previous learning experiences. Meta-learning thus streamlines the adaptation of Transformer models to diverse applications, enhancing their versatility and efficiency.

(a) Neural architecture search:

Neural architecture search (NAS) takes hyperparameter optimization a step further by using machine learning algorithms to explore and identify the most effective model architectures automatically [60]. This approach is particularly valuable when customizing Transformers for unique datasets or complex tasks where traditional architectures may not suffice. NAS can dynamically adjust both the architecture and hyperparameters of a model to optimize its performance. However, the computational demands of NAS are substantial, often requiring significant resources. It is generally recommended to initiate NAS with smaller datasets or scaled-down model variants to manage computational costs effectively while still exploring innovative architectural solutions.

In practical applications, effective optimization of Transformer models depends on balancing model performance and computational efficiency. As the architectural complexity increases—characterized by deeper layers and a higher number of attention heads—the associated computational costs escalate substantially. This necessitates a strategic trade-off, where hyperparameters (Table 2) are meticulously tuned to balance predictive performance against resource consumption. The ultimate objective of hyperparameter optimization is to identify a robust configuration that not only enhances model accuracy but also maintains computational feasibility, ensuring that training and deployment remain within the constraints of available hardware and time budgets. Achieving this balance is critical for sustaining model scalability across diverse tasks and environments. By systematically exploring the hyperparameter space and leveraging advanced optimization strategies, researchers can substantially elevate the operational efficiency and predictive capabilities of Transformer architectures, thereby extending their applicability to increasingly complex problem domains.

2.4. Customized Transformers for time series analysis

Transformers are now a central tool for sequence modeling and are increasingly used in time series forecasting, but time series data—with their strong temporal dependence, changing statistics, and recurring patterns—challenge standard Transformer designs and call for architectures that better reflect temporal dynamics. A key adaptation lies in positional encoding: conventional sinusoidal encodings often struggle to capture seasonality and cyclic trends in domains such as finance and energy, so more expressive temporal encodings are introduced, together with adjusted attention modules and input representations, as summarized in Fig. 4. To handle long sequences without excessive computation, efficient attention variants such as sparse, windowed, or dilated attention are adopted, allowing the model to retain long-range dependencies while scaling to practical data sizes. The architecture is also extended to include relevant external variables (e.g., weather or macroeconomic indicators), which are often crucial for accurate forecasts, and is trained with robust strategies—such as regularization, dropout in attention layers, and curriculum learning—to improve generalization and limit overfitting on noisy, dynamic data. On top of these design choices, a rigorous evaluation framework is needed to assess such customized Transformers for time series forecasting: a diverse set of metrics (Table 3) is used to measure accuracy, robustness, and generalization, providing a multi-faceted view of model behavior under different forecasting conditions. This evaluation offers concrete diagnostic feedback on model design and stability, and directly guides further refinements in temporal encoding, attention, feature integration, and training procedures, so that through iterative improvement, these customized Transformers achieve higher accuracy, better adaptability, and greater robustness across a wide range of temporal forecasting tasks.

2.5. Learning paradigms in Transformers for time-series analysis

Transformer models demonstrate exceptional adaptability and robustness across a spectrum of learning paradigms, including

Table 2

Summary of representative methods for Transformer hyperparameter optimization.

Method	Advantages	Limitations
Grid and random search	Simple to implement; effective for low-dimensional or well-bounded search spaces.	Inefficient in high-dimensional spaces; computational cost scales poorly
Bayesian optimization	Reduces search iterations by modeling the objective function; efficient in complex, expensive settings.	Requires careful model tuning; can be sensitive to surrogate model choice
Gradient-based optimization	Exploits gradient information for fast convergence.	Limited to differentiable objectives; not robust to noisy or discrete spaces
Evolutionary algorithms	Robust to local minima; suitable for discrete and non-differentiable landscapes.	High computational overhead; slow convergence in large search spaces
Population-based training	Simultaneously tunes hyperparameters and weights during training; adapts dynamically.	Complex to parallelize; sensitive to population and mutation design
AutoML	Minimizes manual effort; improves reproducibility and deployment efficiency.	May lack transparency; optimization goals are sometimes opaque
Early stopping and pruning	Reduces overfitting and resource usage; terminates unpromising runs early.	Risk of halting promising models prematurely; may hinder peak performance
Transfer learning for hyperparameters	Leverages knowledge from related tasks to reduce search cost.	Transfer success depends on task similarity; may misguide search
Meta-learning	Learns to generalize tuning strategies across tasks; facilitates few-shot optimization.	Requires extensive prior experimentation; high implementation cost
NAS	Jointly optimizes architecture and hyperparameters; may discover novel model designs.	Computationally intensive; often impractical for large Transformer variants

supervised, unsupervised, semi-supervised, self-supervised, and reinforcement learning (Table 4). Each paradigm addresses distinct challenges in time series analysis—ranging from labeled data scarcity to dynamic decision-making under uncertainty. This flexibility enables Transformers to accommodate varying data regimes and task complexities, facilitating scalable, generalizable, and domain-specific solutions.

(a) Supervised Transformers

In the supervised learning paradigm, Transformers are trained on well-defined input-output pairs, making them highly effective for time-series forecasting tasks. This training allows Transformers to model complex temporal dependencies by minimizing the loss function, which evaluates the discrepancy between predicted and actual outcomes. The ability to capture long-range dependencies has enabled Transformers to outperform other methods in various applications, including stock price predictions and energy demand forecasting. Pretraining Transformers with self-supervised learning further enhances their performance, as demonstrated in protein structure prediction models like trRosettaX-Single, which surpasses AlphaFold2 by leveraging sequence embeddings [61]. Similarly, the MOFormer model for Metal-Organic Framework (MOF) property prediction benefits from self-supervised pretraining, improving prediction accuracy without relying on 3D structural data [62]. Moreover, self-supervised learning approaches in vision Transformers, such as MoCo v3, have shown that large-scale, unlabeled datasets can effectively train models to achieve competitive performance in tasks like object detection [63]. These studies highlight that Transformers, especially when pretrained on large datasets, can achieve state-of-the-art performance in time-series forecasting,

minimizing the need for manually labeled data while maximizing predictive accuracy.

(a) Unsupervised Transformers

Unsupervised learning in Transformers does not rely on labeled outputs. Instead, these models are trained to discern intrinsic patterns and structures from the data itself. In time series analysis, unsupervised Transformers can be applied to tasks such as anomaly detection, where the model identifies unusual patterns or outliers in time series data, or for clustering similar temporal patterns. This paradigm leverages the capacity of the Transformer to process sequential data and extract meaningful information without direct supervision, making it a powerful tool for exploratory data analysis and for gaining initial insights into underlying temporal structures. For example, in medical imaging, Transformers have shown remarkable performance in unsupervised anomaly detection, where they learn the distribution of normal brain data and identify pathological anomalies without the need for labeled data [64]. Similarly, in image registration, unsupervised Transformer-based methods have been employed to establish spatial correspondence between medical images, demonstrating the ability of the model to handle long-range dependencies and improve the accuracy of registration tasks [65,66]. Furthermore, ProtGPT2, an autoregressive Transformer trained on protein sequences, exemplifies the success of unsupervised learning in domains beyond traditional NLP, achieving impressive results in protein design without the need for labeled sequences [67]. These examples underscore the potential of unsupervised Transformers to model complex temporal structures in time series data, offering a powerful tool for uncovering hidden patterns and relationships.

(a) Semi-supervised Transformers

Semi-supervised learning offers significant advantages in time-series analysis, particularly in situations where acquiring comprehensive labeled datasets is challenging or costly. This learning approach utilizes a combination of a small set of labeled data with a substantially larger pool of unlabeled data, thereby enhancing both learning efficiency and model performance without the need for extensive manual labeling. In the realm of time-series data, techniques such as pseudo-labeling prove indispensable. Within this framework, the Transformer model assigns provisional labels to the unlabeled data, which are subsequently treated as actual labels in future training iterations [67]. Additionally, consistency regularization is employed to ensure the stability of the predictions generated by the model despite minor fluctuations or disturbances in the input data [68]. This integrative approach not only improves the precision of forecasts in time-series analysis but also substantially lowers both the costs and labor associated with data preparation [69]. By leveraging semi-supervised learning, it is possible to refine model accuracy while efficiently utilizing available data resources [70].

(a) Self-supervised Transformers

Self-supervised Transformers have demonstrated significant advantages in tasks involving time-series analysis, particularly in situations where acquiring comprehensive labeled datasets is costly or impractical. These models utilize a small set of labeled data in conjunction with a much larger pool of unlabeled data, thereby enhancing learning efficiency and model performance without requiring extensive manual labeling [71]. Techniques such as pseudo-labeling are integral to this framework, where a Transformer model assigns provisional labels to the unlabeled data, which are subsequently treated as actual labels in future training iterations. This process is further enhanced by consistency regularization, which enforces the stability of model predictions under small perturbations or variations in the input data. By incorporating

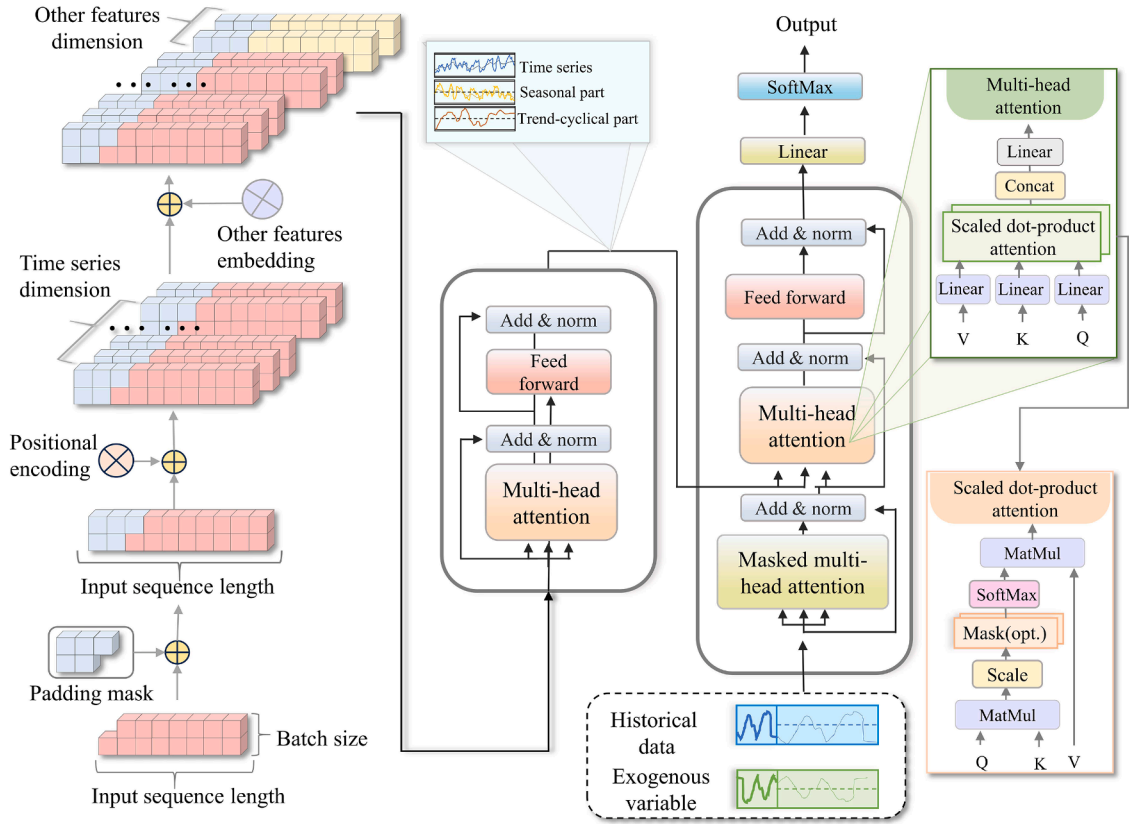


Fig. 4. Framework of Transformer networks for time series analysis and prediction.

Table 3
Evaluation metrics for time series forecasting and their focus.

Metrics	Description	Focus of evaluation
Mean absolute scaled error (MASE)	Scale-independent metric comparing accuracy to a naive baseline. Useful across datasets with different units.	Relative improvement over naive benchmark; scale robustness
Mean absolute percentage error (MAPE)	Error expressed as a percentage of actual values. Common in business forecasting.	Interpretability for stakeholders; relative error magnitude
Weighted mean absolute percentage error (wMAPE)	Weighted extension of MAPE accounting for variable importance.	Error weighted by business significance (e.g., sales volume)
Symmetric mean absolute percentage error (sMAPE)	Symmetric adjustment of MAPE to reduce bias near zero.	Handling low-value forecasts; mitigating denominator asymmetry
Root mean squared logarithmic error (RMSLE)	Emphasizes log differences between predicted and actual values.	Penalizing underestimation; controlling impact of outliers
Theil's U statistic	Compares model error to naive baseline using squared errors.	Relative error ratio over time; model benchmarking
Diebold-Mariano test	Hypothesis test for comparing forecast accuracy of two models.	Statistical significance of performance differences
Quantile loss	Penalizes deviation from target quantiles in probabilistic forecasts.	Calibration of forecast uncertainty across distribution tails

self-supervised learning strategies, models can effectively utilize large volumes of unlabeled data, thereby improving forecasting accuracy while simultaneously reducing the reliance on labor-intensive and costly

data annotation procedures. By capturing both local and global dependencies, self-supervised Transformers excel in tasks like object segmentation and discovery, where they surpass traditional supervised learning approaches that rely heavily on labeled datasets [72,73]. The combination of Transformer models with self-supervised learning has proven to be an efficient and scalable solution, enabling models to discover meaningful patterns from vast, unlabeled datasets.

(a) Reinforcement learning Transformers

Transformers are highly effective in reinforcement learning (RL), particularly for tasks involving sequential decision-making in dynamic environments. Their ability to model long-range dependencies and handle high-dimensional state spaces makes them well-suited for applications like robotic manipulation, autonomous driving, and navigation [74]. In goal-driven tasks, such as autonomous navigation, Transformers improve data efficiency by integrating goal information with scene representations. This enables the model to focus on goal-relevant features, enhancing decision-making and leading to superior performance [75]. Additionally, in multi-agent RL scenarios, such as those in EV charging systems, Transformers enable decentralized coordination among agents, ensuring efficient decision-making with minimal communication costs and preserving privacy [76,77]. This combination of capabilities allows Transformers to navigate complex environments effectively, optimizing both individual and collective agent performance.

2.6. Transformer variants for time series analysis

The evolution of Transformer-based time-series modelling can be delineated into four successive stages—each characterised by targeted architectural refinements that progressively expand the representational capacity for complex temporal signals—as exemplified by the

Table 4

Taxonomy of Transformers by supervision paradigm in time series analysis.

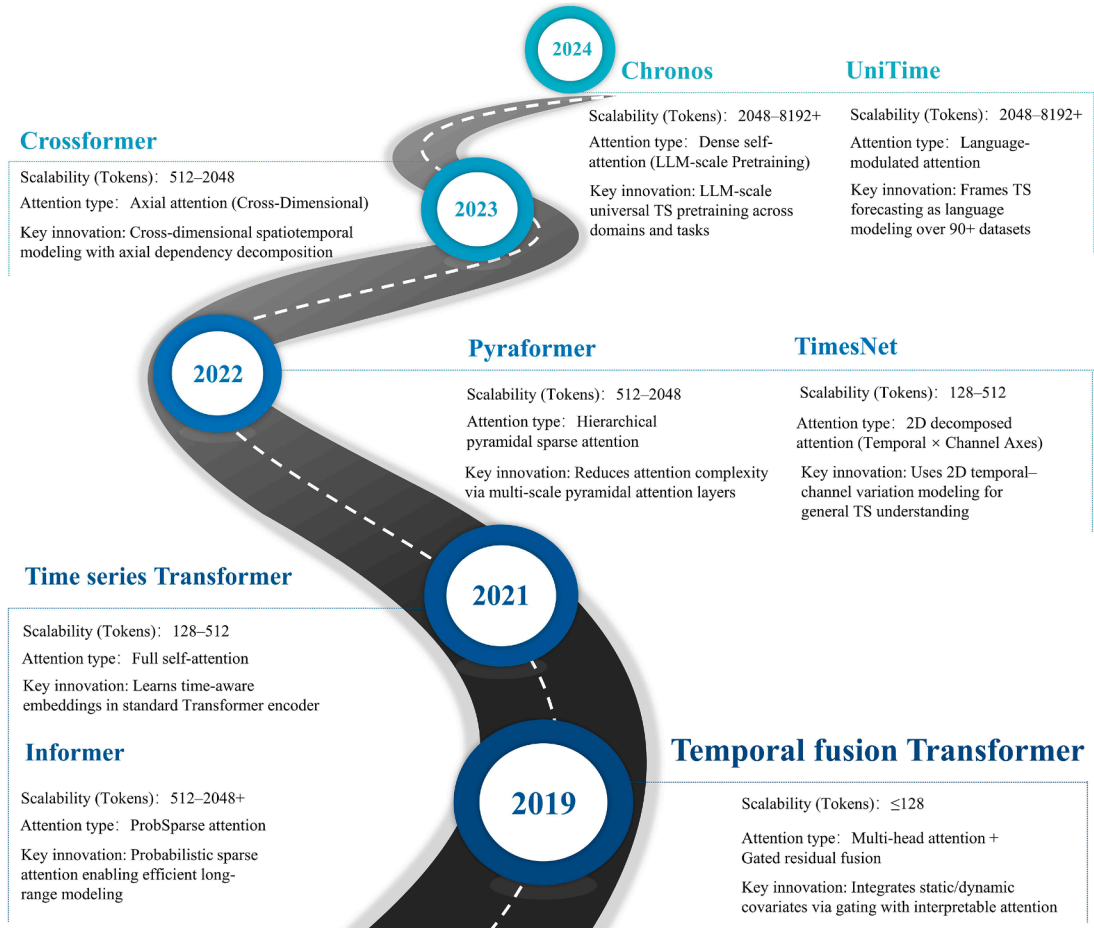
Supervision Type	Definition	Strength	Applications	Advantages / Limitations
Supervised Transformers	Trained on labeled time-series data where both inputs and corresponding outputs are provided.	High predictive accuracy when labeled data is available.	Stock price forecasting in finance; patient outcome prediction in clinical records	High accuracy with rich labels, but label-hungry and fragile under distribution shift.
Unsupervised Transformers	Learn patterns from unlabeled sequences by modeling internal structure or distribution.	Enables learning from abundant unlabeled data.	Clustering customer behaviors in e-commerce; anomaly detection in sensor networks	Learn broad representations from raw data, but weakly aligned with forecasting objectives.
Semi-supervised Transformers	Trained using a small set of labeled data along with a larger unlabeled dataset.	Balances generalization and label efficiency.	Weather forecasting using sparse; labeled stations augmented with remote sensing data	Improve low-label performance using unlabeled data, but risk propagating noisy pseudo-labels.
Self-supervised Transformers	Learn by solving auxiliary tasks, such as predicting masked or future segments within the input.	Captures intrinsic temporal dependencies without labels.	Next-step prediction in multivariate time series; representation learning in wearable sensor data	Enable scalable pretraining and transfer, but add compute cost and task-mismatch risk.
Reinforcement learning Transformers	Applied as policy or value networks in sequential decision-making environments.	Effective in learning long-horizon strategies from feedback signals.	Robotic control optimization; decision-making in autonomous driving and resource allocation	Directly optimize decision reward, but require costly interaction and are hard to train stably.

representative models listed in Fig. 5. The trajectory begins with the TFT, which embeds a Gate Residual Network and a Variable Selection Network within an encoder–decoder backbone. By injecting static and dynamic covariates into the feature stream through gated mechanisms and outputting interpretable variable-attribution scores via multi-head attention, TFT simultaneously achieves reliable multi-step forecasting and transparency demanded by industrial applications [78].

As sequence length and data volume increased, scholarly focus quickly pivoted towards efficient long-horizon modelling. Informer leverages ProbSparse Attention to retain only the highest-entropy queries, reducing the complexity of self-attention from $O(L^2)$ to $O(L\log L)$;

coupled with a self-distilling encoder and a generative decoder, the model sustains linear resource usage even with inputs exceeding one thousand time steps [79]. Contemporaneously, the Transformer-based TST framework incorporates learnable time-shift positional encoding within a dense self-attention architecture and exhibits superior few-shot transferability under an unsupervised pre-train–fine-tune regime, thereby underscoring the viability of full attention in data-scarce scenarios [80].

Once the challenge of long-range dependency modelling had been mitigated, the research emphasis shifted to multi-scale structure and cross-dimensional coupling. Pyraformer constructs a hierarchical pyra-

**Fig. 5.** Evolutionary roadmap of Transformer-based time-series models (2019 – 2024).

midial sparse graph that recursively segments and aggregates sequences, compressing the maximum dependency path to a constant bound and enabling near-linear complexity for extremely long inputs [81]. TimeSNet reshapes one-dimensional time series into a two-dimensional time $\times$ channel tensor and employs a convolution–attention hybrid module to unify forecasting, imputation, classification and anomaly detection, thereby demonstrating architectural unification alongside task diversification [82].

The most recent stage centres on explicit cross-dimensional interaction and the emergence of foundation-model paradigms. Crossformer employs a dual-axial two-stage attention mechanism to model time–variable interactions explicitly, pushing the performance envelope on multivariate benchmarks [83]. Chronos discretises continuous sequences into a fixed vocabulary of tokens and directly re-uses the pre-training objectives of the T5/T0 family, delivering systematic zero-shot gains across forty-two public datasets [84]. Building upon this, UniTime introduces textual domain instructions, treating heterogeneous sequences as a bi-modal input of language + time series and applying mask-guided curriculum learning to balance cross-domain convergence, thereby supporting unified training and inference over more than ninety datasets [85]. Consequently, Transformer-based time-series models have progressed from interpretable, task-specific forecasters to cross-domain foundation models; future research is expected to converge on hybrid sparse/low-rank approximations, edge-oriented compression and multi-task consistent pre-training to further extend their practical frontier.

3. Transformer for time-series forecasting: case studies

The Transformer architecture, renowned for its innovative self-attention mechanism, has demonstrated exceptional capabilities in modeling long-range dependencies, facilitating its transition from

natural language processing to time-series analysis. Compared with traditional recurrent and convolutional neural networks, Transformers provide superior performance in handling multivariate inputs, managing extended temporal sequences, and capturing complex nonlinear dynamics, which together enhance predictive accuracy. Due to their efficiency, scalability, and flexibility, Transformer-based models have been widely applied across diverse fields such as finance, healthcare, energy systems, and transportation, where accurate time-series forecasting plays a pivotal role. As shown in Fig. 6, Transformer-based models are structured in layers, incorporating core components, architectural variants, learning paradigms, and application domains. This review provides a systematic examination of these advancements, emphasizing the potential of Transformer architectures to enhance forecasting accuracy, adapt to dynamic and non-stationary environments, and integrate multimodal data sources. The objective is to present a comprehensive perspective that informs future research directions and guides practical implementations of Transformer-based methods in time-series analysis.

3.1. Finance and stock market analysis

The intricate dynamics of stock markets, marked by volatile fluctuations and complex patterns, have driven growing interest in applying advanced machine learning models, particularly transformers, for precise forecasting. Table 5 summarizes key transformer-based models for stock market prediction, highlighting their capability to capture market complexities and enhance forecasting accuracy. The expanding body of research in this domain underscores the potential of transformers to significantly enhance predictive accuracy, offering a novel and effective approach to navigating the inherent uncertainties of financial markets [86]. A seminal study on the Turkish stock market (BIST100) highlights the promising integration of various deep learning methodologies,



Fig. 6. Multilayered Transformer model framework for time-series forecasting.

including CNNs, RNNs, LSTMs, and particularly bidirectional encoder representations from Transformers (BERT), for the classification of stock direction. This pioneering approach, recognized as the first of its kind, achieved an impressive 96.26 % accuracy, illustrating the power of combining diverse deep learning models with the word embedding model of BERT [87]. Similarly, transformers have emerged as transformative tools for stock price prediction, with one study leveraging self-attention mechanisms within transformer networks to forecast the closing prices of the next day on the Saudi stock exchange with over 90 % accuracy. This work demonstrates the unparalleled capability of transformers to handle high volatility in financial time-series data, marking a significant advancement in the application of deep learning techniques in finance [88].

Subsequent developments in this field have focused on addressing the challenge of long-sequence prediction in multivariate time-series (MTS) data, which requires sophisticated models capable of capturing both periodic and non-periodic temporal patterns. The DST2V-TRANSFORMER architecture represents a notable breakthrough, incorporating techniques such as moving average smoothing and time embedding to enhance the efficacy of transformers in time-series forecasting. The state-of-the-art performance of this model is evidenced by a remarkable 49 % improvement in error rates across seven MTS datasets within the financial sector [89]. The versatility of transformer-based models is further demonstrated by their application in predicting stock prices over extended timeframes, including weeks and months. For instance, in the context of the Dhaka stock exchange (DSE), these models have proven effective not only for regression-based price predictions but also for binary classification tasks, such as predicting price movements. This adaptability is especially pertinent in emerging markets like Bangladesh, where retail investors depend heavily on stock market investments. Furthermore, a comparative study revealed that transformer models, leveraging the multi-head attention mechanism, outperformed traditional deep learning models in both predictive accuracy and net value analysis, setting a new benchmark in financial forecasting [90].

The continued evolution of transformer-based models in stock market prediction reaffirms their growing potential while also highlighting the increasing sophistication of their applications. One such advancement is the TEANet model, which integrates historical stock prices with textual data, such as social media reviews, to predict stock movements. By incorporating multihead and temporal attention mechanisms, the TEANet model excels at capturing complex relationships between stock prices and textual data, demonstrating its effectiveness in real-world applications. This innovative approach has been rigorously validated through extensive experiments, underscoring its potential to transform stock movement prediction by seamlessly blending numerical and textual information [91]. In addition, a comprehensive study comparing transformers with traditional deep learning models in stock market prediction demonstrated that transformers outperformed conventional methods not only in prediction accuracy but also in net value analysis.

The multi-head attention mechanism was identified as a key driver behind this elevated performance, effectively extracting salient features from noisy financial time-series data [92]. As research in this field progresses, the fusion of transformers with complementary technologies, such as natural language processing and sentiment analysis, is poised to further enhance the accuracy and depth of stock market forecasting. The future of this research promises to bring profound improvements in predictive capabilities, benefiting investors globally and reinforcing the crucial role of machine learning in the financial sector.

3.2. Healthcare and medical monitoring

With advancements in disease diagnosis and treatment technologies, the medical field increasingly demands precise analysis of complex data, particularly in time-series forecasting and real-time patient monitoring. As shown in Table 6, representative models discussed in this section are summarized. Machine learning models, especially Transformer architectures, have demonstrated considerable efficacy in addressing these needs by capturing intricate dependencies within sequential data. Their strengths are evident in various medical applications, including time-series prediction, EEG signal interpretation, clinical patient data analysis, and arrhythmia classification. A growing body of research has explored and validated the utility of Transformer-based models in enhancing analytical precision and interpretability in medical contexts. For instance, one foundational study introduced a Transformer-based framework tailored for general time-series forecasting, including both univariate and multivariate inputs as well as time series embeddings. Using influenza-like illness (ILI) as a case study, the model outperformed traditional baselines and showed potential for future adaptation to spatio-temporal data [93]. This general framework laid the groundwork for applications in more specialized and complex medical domains. Building on this, several studies have harnessed Transformer models for critical neurological and clinical prediction tasks. In epilepsy research, EEG signals serve as essential diagnostic data, and one proposed method synergized Fourier-based signal processing with Transformers to identify attention-worthy regions within EEG traces. This hybrid pipeline demonstrated strong classification performance on benchmark datasets, supporting its value for clinical implementation [94]. In the domain of medical claims data, the Claim Pre-Training (Claim-PT) framework applied a two-stage Transformer-based approach—pre-training followed by fine-tuning—on large-scale pediatric data. By learning generalized medical representations, the model effectively addressed downstream tasks with limited data availability, showcasing strong cross-institutional transferability and advancing unsupervised clinical understanding [95]. Likewise, ECG DETR introduced a novel paradigm for arrhythmia detection by treating the problem as an object detection task on continuous ECG signals. Unlike traditional beat-by-beat models, it leveraged Transformers to simultaneously localize and classify heartbeats within unsegmented data, achieving competitive

Table 5
Summary of Transformer-based architectures for stock market forecasting.

Model	Application	Techniques	Dataset	Performance	Reference
BERT	Stock direction classification in BIST100	Combines CNN, RNN, LSTM with BERT embeddings	BIST100	Avg. accuracy: 96.26 %; best: 97.60 % (YBKNN); outperformed CNN, RNN, LSTM	[87]
Transformer w/ self-attention	Volatile next-day stock price forecasting	Self-attention-based sequence modeling	TASI, TBNI	Best MSE: 0.0001 (TASI); MSE range: 0.0013–0.1114 (TBNI); robust MAE, MAPE, RMSE at batch size = 8	[88]
DST2V-Transformer	Long-horizon multivariate financial forecasting	Moving average + time embedding	7 financial MTS datasets	NRMSE: 0.0274 (Exchange), 0.0606 (Indexes); NMAE: 0.0501, 0.0270; 49 % error reduction vs baselines	[89]
Generic Transformer models	Weekly/monthly price regression & movement classification	Multi-head attention	Dhaka stock exchange	RMSE $\approx$ 0.038 (daily), 0.059 (weekly); MAE $\approx$ 0.031 (daily), 0.047 (weekly); superior to traditional models	[90]
TEANet	Stock movement prediction with price and social sentiment fusion	Multi-head + temporal attention with text-price integration	Stock prices + social media	Avg. accuracy: 64.48 %; Avg. MCC: 0.3534	[91]

Table 6

Transformer-based architectures for healthcare applications.

Model	Application	Techniques	Dataset	Performance	Reference
General Transformer	ILI time-series forecasting	Temporal embeddings for uni/multivariate inputs	CDC FluView ILI	Corr = 0.929, RMSE = 0.605; best at dim=8	[93]
Fourier-Transformer	EEG-based seizure prediction	Fourier + Transformer for EEG attention	CHB-MIT, AES	Sensitivity = 98.46 %, FPR/h = 0.124	[94]
Claim-PT	Pediatric risk modeling from claims	Pretrained Transformer + task-specific fine-tuning	1M+ pediatric claims (Partner for Kids)	AUC: 0.84 (suicide), 0.82 (asthma); +10 % vs baselines	[95]
ECG DETR	ECG arrhythmia detection	Transformer-based heartbeat localization and classification	MIT-BIH, PTB-XL	Acc = 99.49 % (MIT-BIH), 10-fold CV	[96]
Hi-BEHRT	Heart failure prediction from EHR	Hierarchical Transformer with contrastive pretraining	UK CPRD EHR (100,071 patients)	AUROC = 0.93, AUPRC = 0.69; robust on small sets	[97]
BERT + LightGBM	Healthcare cybersecurity	BERT embeddings fused with LightGBM	Edge_IIoTset	Acc = 0.9957, AUC = 0.9996, MCC = 0.9837	[99]

performance on standard datasets such as MIT-BIH [96].

Transformer models have also demonstrated compelling performance in risk prediction and longitudinal patient analysis using electronic health records (EHR). One notable study applied Transformer-based learning to predict heart failure using data from over 100,000 patients in the UK. The model incorporated diagnostic codes, medication history, patient age, and calendar year, offering nuanced insights through ablation studies and uncovering previously underestimated risk factors, thereby enriching traditional expert-driven models [97]. However, the complexity and sequence length of EHR data can challenge conventional architectures. To address this, the Hi-BEHRT model introduced a hierarchical Transformer framework that expands receptive fields and captures long-term dependencies. When evaluated on a multimodal EHR dataset, Hi-BEHRT surpassed existing models in predictive accuracy and demonstrated robustness even on smaller datasets via an end-to-end contrastive pre-training strategy [98]. Together, these studies affirm the strength of Transformer models in not only detecting disease risk but also modeling extensive patient trajectories with greater fidelity than earlier approaches.

Beyond diagnostics and prognostics, Transformers are increasingly applied to address cybersecurity vulnerabilities in healthcare systems. As medical infrastructures become more digitized, they face growing risks from cyberattacks targeting patient data and operational continuity. One recent approach combined an optimized LightGBM model with a BERT-based Transformer to form a dual-layer defense: an intrusion detection module for networked devices and a malware detection component for clinical endpoints. Trained on heterogeneous datasets, the system achieved high accuracy and resilience, demonstrating the potential of hybrid AI architectures for robust healthcare security [99]. These developments illustrate the breadth of Transformer applications across the healthcare continuum—from time-series modeling and decision support to institutional data representation and cyber defense. Ongoing innovation at this intersection is likely to yield more accurate, interpretable, and secure AI solutions for emerging clinical and operational challenges.

3.3. Energy storage devices diagnostics

Accurate estimation of the performance of energy storage devices—particularly the state of lithium-ion batteries—is essential for ensuring safety, reliability, and operational efficiency, given their widespread deployment in electronic devices, electric vehicles, and grid-scale storage systems. Traditional model-based approaches, such as equivalent circuit models and physics-informed methods, often struggle to capture the nonlinear degradation behaviors and multiscale temporal dynamics inherent in real-world battery operations. These limitations restrict their applicability under complex and variable operating conditions. As summarized in Table 7, such challenges have driven growing interest in data-driven approaches, among which Transformer-based architectures have shown exceptional promise in modeling long-range dependencies and intricate temporal correlations [100].

Recently, various Transformer variants have introduced distinct architectural innovations across a broad spectrum of battery diagnostic and prediction tasks—including charge state, health, lifetime, and safety estimation [101]. One study enhanced the Transformer with adaptive sampling, dimensional transformation layers, and trainable regression tokens, significantly improving state-of-health (SOH) estimation accuracy while reducing training time across multiple battery datasets [102]. Another proposed a self-supervised Transformer framework that achieved RMSE below 1.9 % across different chemistries and ambient temperatures, demonstrating strong generalization under partially observed data regimes [103]. To address the issue of signal degradation caused by over-denoising, a time-series Transformer incorporating de-stationarization and inception modules was developed, outperforming conventional LSTM and GRU models in statistical learning tasks [104]. Further advances include a generative Transformer-CNN architecture for state-of-charge (SOC) estimation, which achieved R^2 values exceeding 99.5 % under both lab-controlled and real-world conditions [105]. Real-world validation using a Swin Transformer on electric aircraft flight data further confirmed its robustness in the presence of sensor noise and fluctuating discharge profiles [106]. In parallel, hybrid frameworks have emerged that combine convolutional feature extractors with Transformer encoders to jointly capture local degradation patterns and global temporal structures. For instance, a model integrating 1D convolutional layers with a ProbSparse self-attention mechanism achieved an RMSE below 0.7 % and R^2 above 99 % on LFP and NCA batteries, validating its robustness and transferability [107]. Building on this line of work, an intelligent domain-generalized framework for second-life EV batteries coupled a Transformer backbone with few-shot transfer learning and Bayesian optimization, enabling accurate SOH estimation under incomplete cycling data and across heterogeneous SOH domains [108].

Beyond full Transformer architectures, related studies have investigated CNN-attention hybrids that employ selective attention mechanisms without implementing the complete Transformer pipeline. In the context of fast charging—often characterized by signal distortion and limited observability—such a hybrid model achieved SOH estimation with RMSE and MAPE maintained within 5 %, using only partial SOC intervals [109]. To further improve performance under heterogeneous and noisy operating conditions, attention-based models have been enhanced with auxiliary learning strategies and specialized training regimes. For example, FlashAttention-based convolutional fusion networks were introduced to enable resource-efficient battery capacity estimation [110]. More recently, a Predictive Pre-trained Transformer (PPT)—inspired by the Generative Pre-trained Transformer (GPT)—was developed [111]. This model incorporates a domain-adaptive pretraining strategy that integrates unlabeled data with supervised fine-tuning, offering a scalable solution for real-time diagnostics under sparse labeling and irregular sampling conditions.

Collectively, these advancements underscore the transformative role of Transformer architectures in battery diagnostics. Their ability to integrate global and local feature extraction, adapt to multimodal and

Table 7

Transformer-based methods for battery diagnostics.

Model	Application domain	Techniques	Dataset	Performance	Reference
CNN-Transformer	SOH estimation in EV battery packs	CNN + Transformer with PCA and feature scaling	NASA battery dataset	MAE, MAPE, RMSE all < 0.55 %	[100]
Vision Transformer	SOH estimation across multi-cycle datasets	Transformer with adaptive sampling and regression token	NASA and University of Maryland datasets	Accuracy > 0.99 across test sets	[102]
SSL-Transformer	SOC estimation under varying ambient conditions	Self-supervised Transformer (no feature engineering)	Multiple Li-ion cells, various chemistries and temperatures	RMSE = 0.90 %, MAE = 0.44 %, generalizes to unseen chemistries	[103]
DendiNet (Time Series Transformer)	SOC estimation under over-denoised and non-stationary signals	De-stationary and Inception modules with Time Series Transformer	Public and ENERO C proprietary datasets	RMSE reduction by 6.72- 7.17 % vs. baselines	[104]
Transformer+CNN	SOC estimation from real-road and lab datasets	Transformer generator + CNN critic	Experimental + real-world road datasets	$R^2 > 99.5$ % across datasets	[105]
Swin Transformer	SOC estimation for electric aircraft battery systems	Swin Transformer with interpolation and noise mitigation	Real flight data from electric aircraft	Acc = 98.5 %, avg. rel. error ≈ 1.5 % (two-seat); Acc > 97 %, avg. rel. error < 3 % (four-seat);	[106]
Specialized Convolutional Transformer Learning	SOH estimation across battery chemistries	Domain-specific CNN + Transformer encoder; transfer learning	NASA, Oxford, and other public datasets	RMSE = 0.247 %, $R^2 = 99.8$ %, WMAPE = 0.223 %, MAE = 0.202 % for SOH estimation of LFP batteries	[107]
Intelligent Domain-Generalized SOH Estimation	Second-life EV battery SOH under incomplete cycling and domain shifts	ResNet-Transformer-KAN; FSTL-based domain alignment; Bayesian hyperparameter optimization	UCSD-XJTU-Nissan; IECON-2022	MSE 0.114 %, MAE 0.852 %; IECON-2022: MSE 0.273 %, MAE 1.861 %	[108]
FastCharge Former	SOH estimation under fast-charging protocols	CNN-based multi-fusion with self-attention; deep transfer learning	Oxford EV battery dataset (146,074 cycles)	MAPE = 0.389 %, RMSE = 0.479 %, $R^2 = 97.7$ % for SOH estimation of LIBs	[109]
FlashFormer	Battery capacity estimation with low resources	Lightweight CNN encoder + FlashAttention Transformer fusion	Proprietary + public EV datasets	RMSE = 0.109 %, $R^2 = 0.998$, MAPE = 0.096 % for SOH estimation of LFP cells	[110]
Predictive Pretrained Transformer	Real-time SOH and fault diagnostics	Transformer backbone with pretraining and supervised fine-tuning	Real-world field data from EVs	RMSE = 0.235 %, $R^2 = 98.9$ %, WMAPE = 0.190 %, MAE = 0.183 %	[111]

cross-domain datasets, and generalize across battery chemistries and operating environments positions them as a foundational tool for next-generation battery management systems. Moving forward, research should focus on enhancing interpretability, computational scalability, and lifelong learning capacity to support real-time, embedded implementations in safety-critical energy storage platforms.

3.4. Transportation and traffic prediction

Traffic flow forecasting is undergoing a transformative shift through the integration of methodologies from NLP, particularly with the adaptation of Transformer architectures. As shown in Table 8, representative models discussed in this section are summarized, illustrating their approaches and effectiveness in capturing complex temporal and spatial dependencies within traffic systems. Originally developed for linguistic sequence modeling, these architectures have been repurposed to capture complex temporal and spatial dependencies within transportation systems. Recent works have demonstrated how attention mechanisms can effectively model non-linear traffic dynamics. For example, the traffic Transformer model [112] introduces a hybrid design that incorporates temporal segment analysis and graph convolution to process dynamic traffic flows, while maintaining an emphasis on sequential attention. Similarly, Bi-STAT [113] applies spatial and temporal adaptive components within an encoder-decoder framework, supported by a dynamic halting module that adaptively regulates sequence length based on traffic conditions. TrafficBERT [114] follows a pretraining strategy analogous to language models, enabling generalized forecasting across road networks by leveraging only weekday and traffic speed data, without relying on auxiliary features. In parallel, the meta graph Transformer (MGT) [115] combines multiple attention layers—temporal, spatial, and embedding-based—while utilizing heterogeneous spatial-temporal embeddings to accommodate variable

traffic conditions. Collectively, these studies highlight the adaptability of Transformer-based frameworks to diverse traffic forecasting scenarios.

Advancements in travel time estimation and traffic demand modeling have further demonstrated the applicability of Transformer architectures to complex urban prediction tasks. STGNN-TTE [116], a model combining spatial-temporal graph convolution with Transformer modules, enables simultaneous prediction of link-level and route-level travel times. By capturing both long-term dependencies and local fluctuations, it addresses one of the major shortcomings of prior models. In the domain of demand forecasting, DSTGNN [117] adopts a dynamic graph construction approach based on information entropy and integrates an inhomogeneous Poisson process to simulate variable traffic intensities. The use of a deep convolutional neural network (DCNN) alongside enhanced attention mechanisms improves spatial dependency modeling under uncertain conditions. For traffic signal control, a reinforcement learning-based system has been proposed, wherein Transformer modules are embedded in a fog computing environment to control signal timing across multiple intersections [118]. This approach leverages asynchronous DRL algorithms such as A2C and PPO for phase optimization, resulting in significant performance improvements across simulated and real-world traffic networks. Meanwhile, Lastjormer [119] introduces a refined attention mechanism by jointly modeling spatial and temporal dependencies across sensor networks, employing linear and convolution-based attention variants to reduce computational complexity while enhancing accuracy.

Beyond forecasting and control, Transformer models have been extended to novel applications in transportation cybersecurity, congestion analysis, and mobile traffic prediction. FED-IDS [120], a federated deep learning framework, integrates a context-aware Transformer to analyze distributed traffic data for intrusion detection. Blockchain validation ensures secure and decentralized learning across edge

Table 8

Transformer-based methods for traffic forecasting, control, and vehicular network security.

Model	Application domain	Techniques	Dataset	Performance	Reference
Traffic Transformer	Highway traffic speed forecasting	Temporal segments + GCN + attention	METR-LA, PEMS-BAY	MAE = 2.43, RMSE = 4.77, MAPE = 6.57 % (15 min, METR-LA)	[112]
Bi-STAT	Urban traffic prediction with sequence control	Dynamic halting + adaptive spatial-temporal encoder	PeMSD3-D8	Avg RMSE = 4.25, best across 12 steps (PeMSD7)	[113]
TrafficBERT	Long-range traffic flow forecasting with BERT pretraining	Pretrained BERT on weekday-speed data	PeMS-Bay, METR-LA	MAE = 3.53, RMSE = 5.72, MAPE = 7.72 % (PeMS-Bay)	[114]
MGT	Graph-based traffic prediction under spatial heterogeneity	Meta-graph spatial/temporal/embedding attention	PeMSD4, METR-LA	RMSE = 3.96 (PeMSD4), better under spatial heterogeneity	[115]
STGNN-TTE	Travel time estimation at route and link levels	ST-GCN + gated Transformer fusion	Chengdu1/2, Porto	RMSE = 88.03 (Chengdu1), MAE = 57.45 (multi-scale travel time)	[116]
DSTGNN	Urban traffic demand prediction	Poisson-based dynamic graph + attention	TaxiNYC, Bluebike	MAE = 2.91, RMSE = 5.14 (TaxiNYC), strong under demand fluctuation	[117]
Transformer + DRL	Traffic signal control via DRL-Transformer	Transformer + A2C/PPO in fog architecture	SUMO, city simulation	Reduced avg delay by 18 %, throughput ↑ 21 % (simulated intersections)	[118]
Lastjormer	Efficient traffic forecasting with reduced complexity	Linear + convolutional attention	Sensor networks	MAE = 2.78, 30 % faster inference vs. ST-GCN with same accuracy	[119]
FED-IDS	Cyberattack detection in vehicle networks	Federated Transformer + blockchain	Edge_IloTset	Acc = 0.9957, AUC = 0.9996, MCC = 0.9837 (Edge_IloTset)	[120]
STTF	Congestion prediction with spatial-temporal fusion	Road speed + topological fusion	Congestion datasets	RMSE = 5.01, MAE = 3.11, improves congestion prediction	[121]
STD-Net	Urban mobile traffic forecasting	Hierarchical Transformer with downsampling	Mobile traffic logs	MAE = 0.091 (Porto), RMSE = 0.143, effective for 6 G network allocation	[122]

devices. STTF [121] focuses on congestion prediction by incorporating road speed, topological constraints, and spatial-temporal interactions through a Transformer-based structure. For mobile network traffic forecasting, STD-Net [122] utilizes spatial-temporal Transformers and hierarchical down sampling to extract high-resolution features from dense urban datasets, thereby supporting resource allocation in future 6 G systems. A recent comprehensive review classifies traffic prediction models into CNN-, GCN-, and Transformer-based categories, evaluating them across benchmarks and analyzing their architectural strategies [123]. Results suggest that models combining sequence-to-sequence structures with graph learning and attention mechanisms consistently achieve superior performance, especially under dynamic spatial-temporal settings. These findings indicate an ongoing trend toward multi-component hybrid models capable of adapting to a broad spectrum of traffic-related challenges.

As Transformer-based methodologies continue to evolve, they are poised to redefine the boundaries of urban mobility forecasting. Their ability to model both macro-level patterns and fine-grained temporal shifts makes them especially suited for intelligent transportation systems. However, key limitations remain, including the computational intensity of large-scale models, the need for interpretability in safety-critical environments, and challenges associated with sparse or noisy urban sensor data. Addressing these challenges necessitates the development of more efficient Transformer variants, advanced transfer learning strategies, and robust frameworks for real-time deployment. As demonstrated in the reviewed literature, Transformer architectures—when synergistically combined with emerging techniques such as graph-based learning, reinforcement learning, and cloud-edge collaborative computing—constitute a promising foundation for next-generation urban traffic analytics. Ongoing advancements in the design, optimization, and integration of these models are anticipated to play a critical role in enabling scalable, accurate, and resilient traffic forecasting systems within the context of future smart cities.

4. Current challenges

Transformers have markedly advanced time series analysis, offering distinct advantages that support their widespread adoption across domains. These models excel at capturing complex, long-range dependencies, often outperforming conventional machine learning approaches. Powered by self-attention mechanisms, Transformers process entire sequences in parallel, yielding deeper insights into temporal

dynamics. Their flexibility enables the modeling of multivariate time series, facilitating the simultaneous analysis of interdependent data streams—critical for applications such as financial forecasting, energy management, and supply chain logistics. Moreover, their scalability allows efficient processing of large-scale datasets, aligning with the demands of an increasingly data-driven landscape. This section examines the key challenges (Fig. 7) confronting Transformers. While they demonstrate strong generalization and versatility, several practical and theoretical obstacles persist, which can limit their effectiveness in real-world deployments.

4.1. Computational scalability and cost

Transformers, while offering substantial advantages for time series analysis, encounter significant computational challenges that can hinder their practical application [124]. The architecture of these models, particularly when tasked with analyzing extensive time series data, requires a substantial amount of computational power. This necessity stems from their complex self-attention mechanisms, which, although effective in capturing long-range dependencies and processing sequences in parallel, are computationally intensive. The demand for high computational resources is further amplified when handling multivariate time series with high dimensionality. Each additional variable introduces more complexity and lengthens the computation time, which can be prohibitive for real-time applications or scenarios with limited computational infrastructure. Moreover, the training process for Transformers involves adjusting a large number of parameters, a task that not only requires significant computational power but also considerable time, especially when dealing with large datasets [125]. These computational demands pose a challenge, particularly in deploying Transformers in environments where computing power is at a premium. The requirement for extensive resources can limit the adoption of these models in smaller organizations or applications where rapid response times are crucial. Therefore, addressing the computational expense associated with Transformers is essential for broadening their applicability and ensuring that they can deliver their full potential across a wider range of time series analysis tasks.

Evaluating the computational demands of deep learning models is essential for assessing their efficiency, scalability, and feasibility in real-world deployments. A thorough analysis of resource requirements—such as memory consumption, processing time, and hardware dependencies—enables researchers and practitioners to make informed

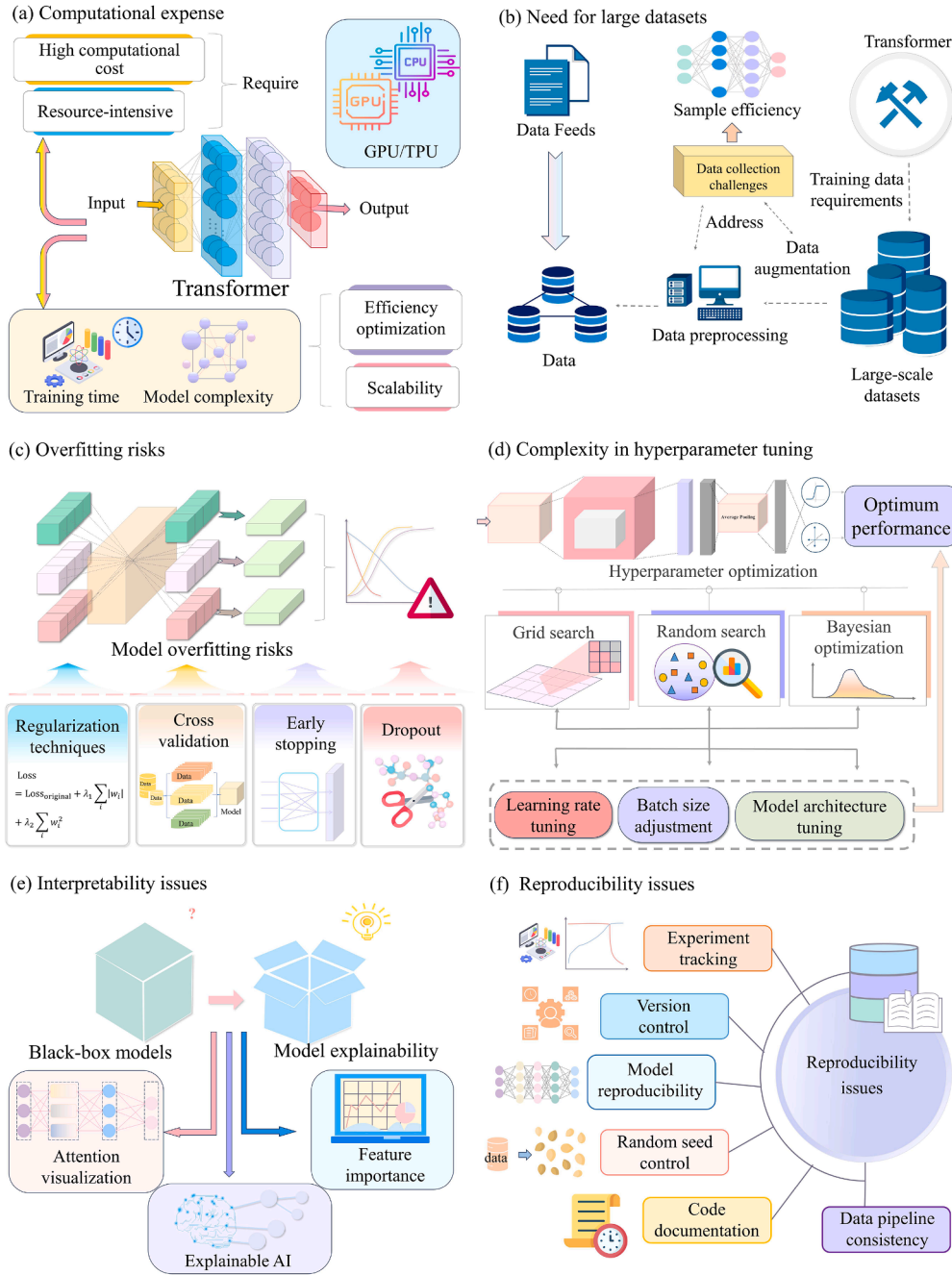


Fig. 7. Key challenges in Transformer-based time series modeling. (a) Computational scalability and cost; (b) Limited labeled data availability; (c) Generalization and overfitting risks; (d) Hyperparameter tuning complexity; (e) Limited interpretability and explainability; (f) Reproducibility and benchmarking challenges.

decisions regarding model selection, architectural trade-offs, and optimization strategies tailored to specific application scenarios. One widely recognized metric is the number of floating-point operations (FLOPs) conducted during execution of the deep neural networks [126]. This measurement acts as a direct indicator of computational complexity and is invaluable for comparing the efficiency of various models. It is especially relevant when processing large datasets or when computational resources are constrained. Memory usage is another important metric [127], indicating the amount of RAM required by a model. This is particularly critical for applications on devices with limited memory capacity, such as mobile phones and embedded systems. Lowering memory requirements helps to make deep learning models more versatile and applicable to a wider range of technologies. In addition, the energy consumption of these models has become an increasingly

significant consideration due to the environmental impacts of computing [128]. Developing models that consume less energy not only promotes sustainability but also improves cost-effectiveness, especially in large-scale implementations. By systematically analyzing these metrics, we can enhance our understanding of the necessary trade-offs involved in the design and deployment of deep learning models. Such insights are essential for advancing the field in a way that balances computational efficiency with practical applicability.

4.2. Data hunger and limited label availability

A central challenge in using Transformers for time series analysis is their strong reliance on large datasets [129,130]. Because Transformers are designed to learn complex patterns and long-range dependencies,

they typically require many training examples to capture the underlying structure and variability in the data. When only limited data are available, their performance can degrade or become unstable. In practice, this data demand is often difficult to meet. Collecting time series data can be costly, subject to strict privacy rules, or limited by access constraints. In sectors such as healthcare and finance, obtaining datasets that are both comprehensive and compliant with regulatory requirements is particularly challenging [131]. Small markets or emerging application areas may also lack sufficient historical records. In addition, building and maintaining large datasets—through collection, cleaning, updating, and storage—adds substantial time, cost, and complexity to projects involving Transformers. These issues highlight the importance of developing methods that either reduce the data requirements of Transformers or allow them to perform reliably with smaller datasets, thereby enabling broader and more practical use in time series analysis and prediction.

4.3. Real-world temporal complexity and data heterogeneity

In real-world settings, the properties of time series data themselves pose a major obstacle to deploying Transformer-based forecasting models. Time series collected in climate science, transportation, neuroscience, and environmental monitoring are inherently spatio-temporal: observations are strongly correlated across both space and time, often violate the assumption of independent and identically distributed samples, and show marked differences across regions and periods. Atluri et al. [132] further demonstrate that such spatio-temporal datasets combine multiple data types (e.g., events, trajectories, fields) and require task-specific instance definitions and similarity measures, while exhibiting non-stationary behavior and regime shifts that call for locally adapted models rather than a single global solution. At the same time, modern forecasting problems involve rich temporal structure—overlapping trends, seasonal patterns, multi-scale dynamics, and irregular sampling—as well as complex interactions among variables in high-dimensional multivariate series. Recent surveys on deep learning for time series note that effective models need to jointly capture temporal correlations, cross-variable dependencies, and changing statistics, often by decomposing data into trend, seasonal, and multi-scale components to make learning more manageable [133]. Lim and Zohren [134] further argue that the diversity of application domains—from weather and climate to healthcare, retail, and finance—has driven the development of many specialized and hybrid architectures, underscoring the difficulty of designing a single model that generalizes across datasets with different sampling rates, noise characteristics, and forecast horizons. For Transformer-based forecasting, this temporal complexity and data heterogeneity translate into a need for domain-aware positional encodings, mechanisms that respect spatio-temporal structure, and adaptive normalization or conditioning schemes that can handle shifting distributions across entities, locations, and time. Without explicitly addressing these factors, even carefully tuned architectures are likely to exhibit brittle performance when transferred from curated benchmarks to operational environments, making real-world temporal complexity and data heterogeneity a central challenge for current and future Transformer models in time series forecasting.

4.4. Generalization and overfitting risks

One of the main challenges Transformers face in time series analysis is their tendency to overfit, particularly when trained on extensive datasets [135]. Overfitting occurs when a model becomes overly specialized in the details and noise of the training data, compromising its performance on new, unseen data. For instance, one study demonstrated that while the naive Repeat method performs poorly in predicting long-term seasonal data, such as electricity consumption, it surprisingly outperforms all Transformers in predicting exchange rates by about 45

% [136]. This discrepancy arises because Transformer-based solutions often overfit to sudden changes in the training data, leading to significant accuracy degradation in trend predictions. This issue underscores the need for careful model selection and tuning to ensure robust performance across different types of time series data. This tendency is pronounced with Transformers, given their deep and sophisticated structures that enable them to detect subtle patterns and variations. The propensity for overfitting is especially problematic in time series analysis, where the predictability of future events often depends on subtle patterns that may not be well-represented in historical data. When a Transformer overfits, it may capture these idiosyncrasies instead of generalizing from broader trends, resulting in forecasts that are less accurate or reliable. Addressing overfitting requires careful tuning of the model, including adjustments to the number of layers, the size of the model, and regularization techniques such as dropout. Implementing validation techniques, such as cross-validation or splitting the time series data into distinct training and validation sets, is also crucial. These strategies help ensure that the Transformer model generalizes well from training data to real-world applications, maintaining robustness and reliability in its predictions.

4.5. Hyperparameter tuning complexity and instability

Hyperparameter tuning presents a significant challenge in the deployment of Transformers for time series analysis. The effectiveness of Transformers is largely dependent on the precise configuration of their hyperparameters, which include settings related to the number of layers, the size of the attention mechanisms, and the learning rate, among others [9]. The complexity of these models, combined with their deep architecture, necessitates a meticulous approach to tuning these parameters to achieve optimal performance. However, identifying the ideal set of hyperparameters is not straightforward and can be a time-consuming and computationally expensive process. This difficulty is compounded by the fact that the optimal hyperparameter settings can vary significantly between different applications and datasets. As a result, practitioners often need to engage in extensive trial-and-error experimentation to tailor model configurations to specific time series forecasting tasks. Moreover, the lack of a standard procedure for hyperparameter optimization in Transformers complicates their broader application across diverse time series problems. To address this challenge, researchers and practitioners have increasingly adopted advanced optimization techniques—such as grid search, random search, Bayesian optimization, and AutoML tools—to systematically explore and evaluate a wide range of hyperparameter combinations. Successfully managing the complexity of hyperparameter tuning is crucial for harnessing the full potential of Transformers in time series analysis and enhancing their accuracy and applicability in real-world forecasting scenarios.

4.6. Limited interpretability and explainability

Despite their impressive performance in time series forecasting, Transformer-based models face a fundamental limitation: poor interpretability. Their architecture relies heavily on deep stacking, multi-head self-attention, and nonlinear transformations, which, while enabling the capture of complex temporal dependencies, also make it extremely difficult to understand how predictions are made [137]. This black-box nature is particularly problematic in time series domains where the input is inherently sequential, multi-dimensional, and often influenced by latent temporal or external factors. Unlike in NLP or CV tasks where token-wise or region-wise attention can be easily visualized, in time series data the entanglement of temporal, static, and external covariates makes direct attribution far less intuitive [138]. Moreover, standard post-hoc interpretability methods such as SHAP or saliency maps often fail to respect the temporal structure of input data, leading to misleading or inconsistent explanations [139]. These challenges are amplified in multivariate or high-dimensional forecasting tasks, where

the interactions across variables and time steps create additional ambiguity in understanding the decision-making process.

Several empirical studies have exposed these interpretability gaps. For example, IHTF highlights that even when attention weights are available, they may only partially reflect true feature influence due to interference from residual and gating mechanisms. TFT attempts to offer variable-level interpretability, but its ability to capture deep temporal interactions remains limited by the separation of static and dynamic input processing. Similarly, in tasks like soil moisture or wind power forecasting, attention visualizations alone cannot explain model behavior when interacting meteorological or geophysical signals are involved [140]. In temporal causal discovery, methods like CausalFormer have shown that relying solely on attention weights overlooks significant contributions from fully connected layers and other model components [141]. This indicates that attention interpretability, while valuable, is neither sufficient nor comprehensive for causal or explanatory understanding. Therefore, the core challenge lies not only in visualizing which time steps are important, but in building models whose internal logic can be systematically and faithfully explained across temporal, variable, and contextual dimensions.

4.7. Reproducibility and benchmarking challenges

Reproducibility presents a considerable challenge in the deployment of Transformers for time series analysis and prediction [142]. This issue arises from variations in model initialization, data preprocessing, and training environments, which can lead to inconsistent outcomes across different implementations. Such variability not only complicates the validation of model effectiveness but also hinders the broader adoption of Transformer-based solutions in critical sectors. Ensuring the reproducibility of results obtained from Transformer models requires rigorous standardization of the development process for the model. This includes clearly documenting the data preprocessing steps, using consistent initialization techniques, and specifying the computational environment in which the models are trained. Additionally, sharing detailed model configurations and the hyperparameters used during training can help other researchers and practitioners replicate the findings. Addressing these reproducibility challenges is essential for advancing the reliability of Transformer applications in time series analysis. By establishing and adhering to strict protocols for model development and reporting, the research community can better assess the true capabilities of these models and facilitate their practical implementation across various domains.

5. Outlook

The future of Transformers in time series analysis and prediction is highly promising, fueled by continuous advancements in model architecture and optimization techniques. These innovations are poised to further enhance the ability of Transformers to manage complex temporal data, thereby expanding their applicability across various industries. These advancements not only improve the efficacy of Transformers in traditional forward prediction tasks but also extend their utility to inverse problems, where discovering underlying physical principles or solving high-dimensional challenges is crucial. The ability of Transformers to uncover hidden patterns and relationships within data has emerged as a major research focus, driven by rapid methodological and application-oriented advances. Subsequent subsections examine these developments (Fig. 8), highlighting their growing influence on time series analysis and prediction across multiple domains.

5.1. Physics and domain-informed Transformer design

The future integration of domain knowledge into Transformer models holds significant promise for advancing time series analysis and prediction. By incorporating domain-specific features and expert

insights, these models can achieve a higher level of context-awareness, leading to more accurate and relevant predictions. This approach allows Transformers to tailor their predictions based on the unique characteristics and requirements of specific fields, such as finance, healthcare, and environmental science. A notable advancement in this area is the concept of domain adaptive intermediate fine-tuning (DAIFT) [143]. DAIFT involves fine-tuning Transformer models on intermediate tasks that are closely related to the target domain before applying them to the primary task. This intermediate fine-tuning helps the model adapt to the specific characteristics and nuances of the domain, enhancing its performance and predictive accuracy. By incorporating DAIFT, Transformer models can better leverage domain-specific knowledge, making their predictions more precise and contextually relevant.

Integrating domain knowledge can also enhance the interpretability of Transformer models, addressing one of the critical challenges currently faced in their deployment. When models incorporate expert knowledge, their decision-making processes become more transparent and aligned with established domain-specific principles. This transparency not only boosts the trust of stakeholders but also facilitates the acceptance and implementation of these advanced models in practical applications. Furthermore, the integration of domain knowledge enables the development of customized solutions that are finely tuned to the nuances of different datasets and predictive tasks. For example, in healthcare, incorporating clinical expertise can help refine models to predict patient outcomes more accurately, while in finance, integrating market theories can improve the forecasting of economic trends. The integration of domain knowledge, supported by techniques such as DAIFT, represents a significant step forward in enhancing the accuracy, interpretability, and applicability of Transformer models in time series analysis. By incorporating domain-specific knowledge from diverse fields, these models are capable of generating more accurate and context-aware predictions, thereby fostering innovation and enhancing operational efficiency across a wide range of industrial applications.

5.2. Hybrid architectures and model fusion

The future of Transformers in time series analysis and prediction is poised for substantial advancement through the integration of multi-model fusion strategies. This approach systematically combines models that specialize in distinct temporal characteristics, constructing a unified framework capable of capturing both short-term fluctuations and long-term dependencies [144]. By leveraging these complementary strengths, multi-model fusion mitigates the inherent limitations of individual models, enhancing predictive accuracy and robustness in complex, non-stationary time series environments. For instance, in hydrological forecasting, the fusion of neural networks with probabilistic and fuzzy logic models has consistently demonstrated superior performance over standalone models, particularly in adapting to variable data regimes.

In addition to enhancing forecasting accuracy, multi-model fusion significantly improves generalizability by mitigating the risk of overfitting to specific data patterns. By aggregating outputs from diverse models, this approach balances individual model biases and promotes robustness, thereby ensuring more reliable performance on previously unseen data. This characteristic is particularly critical in volatile time series domains, where abrupt structural shifts and non-stationary behaviors frequently occur. Moreover, this strategy fosters interpretability by enabling the decomposition of the contribution of each model within the fusion framework. Such transparency elucidates the decision-making process, building trust among stakeholders, especially in domains where explainability is paramount, such as healthcare and finance [145]. Recent studies in epidemic event extraction further support this perspective, demonstrating that multi-model fusion mitigates recognition bias and improves generalization across heterogeneous datasets. As research progresses, the integration of fusion methodologies with Transformer architectures is expected to become a cornerstone of time

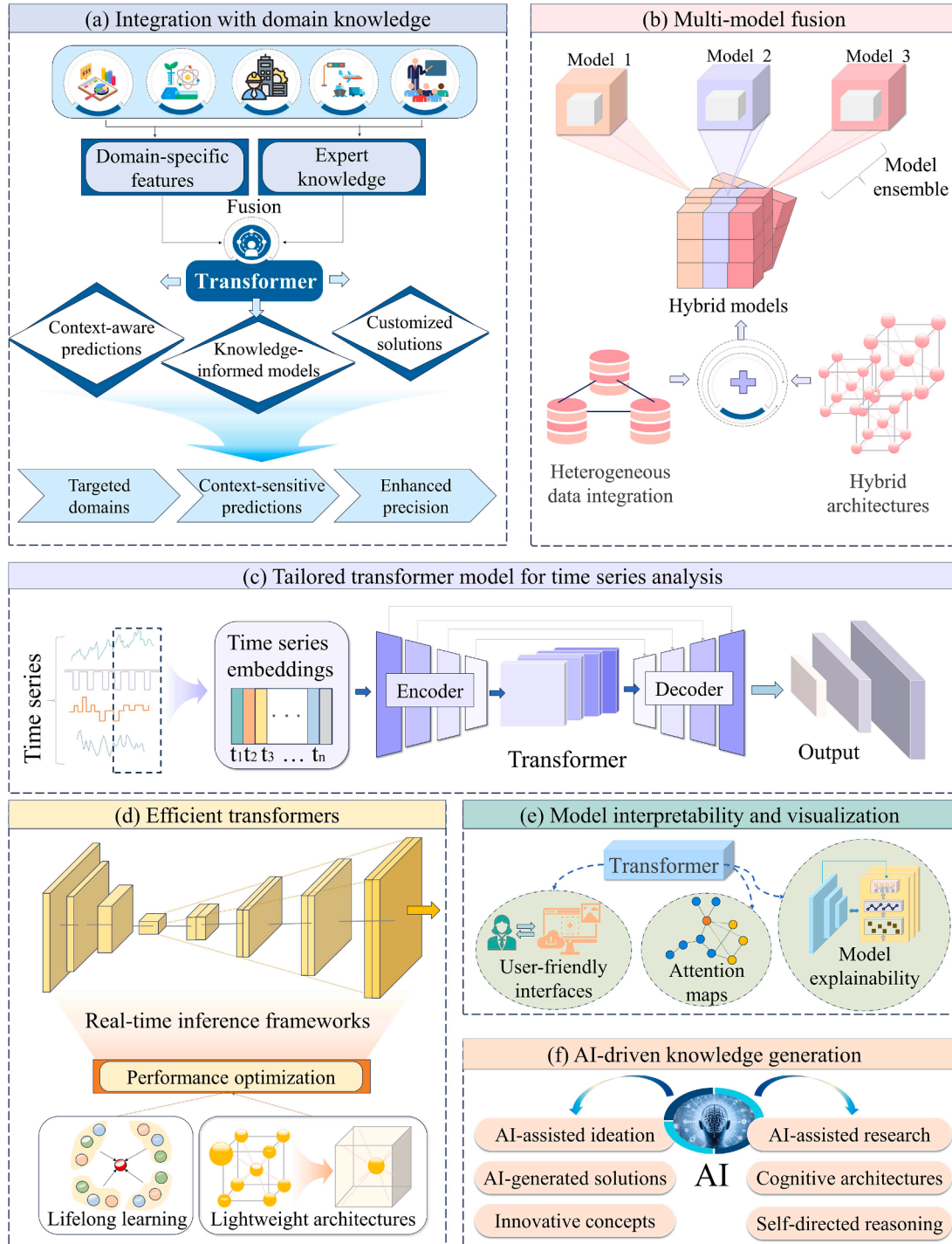


Fig. 8. Advancements of Transformer models in time series analysis.

series forecasting, advancing the accuracy, resilience, and interpretability of predictive systems across diverse application domains.

5.3. Custom Transformer architectures for temporal modeling

Customizing Transformer architectures to address the unique challenges of time series data can improve their performance and applicability across various domains. One key aspect of tailoring Transformers for time series analysis is the incorporation of specialized positional encodings that respect the temporal nature of the data. Unlike traditional positional encodings used in natural language processing, time

series-specific encodings can capture periodicities and other temporal patterns more effectively. This adaptation ensures that the model accurately understands the sequence and relationships within the data, leading to more precise predictions. Furthermore, integrating domain-specific features and expert knowledge into Transformer models can greatly enhance their context-awareness. By incorporating relevant domain insights, these models can provide more accurate and relevant predictions tailored to specific applications such as finance, healthcare, and energy management. This approach not only improves the accuracy but also its interpretability, making it easier for stakeholders to trust and utilize the predictions. Another critical element is the adaptation of

attention mechanisms to better handle the characteristics of time series data. Modifying the self-attention mechanism to focus on relevant time points and ignore noise can improve the efficiency and accuracy of the model. Techniques such as sparse attention or temporal attention mechanisms are particularly effective in managing long sequences and reducing computational overhead. The development of tailored Transformer models for time series analysis represents a significant advancement in this field. By incorporating specialized positional encodings, domain-specific features, and advanced attention mechanisms, these models can achieve higher accuracy and efficiency. As research and innovation continue, tailored Transformers are set to become indispensable tools for time series analysis and prediction, driving progress across multiple industries.

5.4. Resource-efficient Transformers for real-time inference

Transformers, initially developed for processing sequential data in natural language tasks, have emerged as powerful tools in time series analysis and prediction. Their proficiency in detecting patterns across extensive sequences makes them ideal for forecasting applications that require a deep understanding of historical contexts. However, the high computational demands of traditional transformer architectures can be a barrier to their use in real-time applications.

Recent technological advancements have focused on creating more efficient transformer models that preserve accuracy while minimizing computational overhead. These advancements include the development of techniques such as model pruning [146,147] where redundant parts of the network are eliminated without loss of performance, and quantization, which reduces the precision of the calculations to speed up processing. Another significant innovation is the adoption of sparsity within the transformer layers [148,149], which involves activating only relevant parts of the network based on the specific data being processed. These refined models are designed to operate within the stringent time constraints of real-time forecasting environments. By integrating these efficient transformers, forecasters can access in-depth insights more swiftly, eliminating the delays typically associated with complex computational processes. This combination of reduced latency and maintained analytical power facilitates the creation of dynamic and responsive forecasting systems, which are especially vital in fields such as finance, meteorology, and logistics, where rapid data analysis is crucial. The implementation of these streamlined transformers marks a critical step forward in the evolution of forecasting technology, offering both improved speed and decision-making capabilities. This technological progression enables businesses and organizations to react more adeptly to changing conditions, harnessing real-time data for better, more informed strategic planning and operations.

5.5. Interpretability and spatiotemporal visualization

Transformers excel in modeling complex dependencies for time series analysis, yet their broader application depends on enhancing interpretability and result visualization. The self-attention mechanism inherently supports traceability, enabling direct mapping between inputs and outputs, which contrasts with the opacity of traditional deep networks [150]. Leveraging this, effective interpretability depends on going beyond technical transparency to meet the cognitive needs of diverse stakeholders, ensuring explanations are understandable, context-relevant, and capable of fostering trust [151]. Embedding explainable artificial intelligence (XAI) principles into Transformer models not only clarifies internal processes but also aligns model outputs with the expectations of users, regulators, and decision-makers.

Complementing interpretability, robust visualization tools are vital for transforming complex outputs into actionable insights. Dynamic techniques such as attention heatmaps, saliency maps, and counterfactual visualizations reveal feature contributions and temporal patterns, making model behaviors more transparent and accessible [152]. When

integrated with domain knowledge, these tools further enhance decision-making, as shown in applications across healthcare and finance [153]. Together, interpretability and visualization amplify the usability and reliability of Transformers, ensuring not only predictive accuracy but also the transparency necessary for informed, trustworthy decisions in critical sectors.

5.6. Human-AI co-creation and generative intelligence

The concept of AI-driven thought creation represents a pivotal advancement in artificial intelligence, fundamentally transforming our approach to problem-solving and idea generation. This innovation is exemplified by the development of the "Algorithm of Thoughts" (AoT) methodology [154]. Moving away from traditional methods, AoT introduces a dynamic, algorithm-based form of reasoning within, marking a significant shift towards processes that mimic human cognitive functions.

Historically, LLMs were designed to provide direct answers, mimicking a rapid decision-making process akin to human reflexive thinking. However, modern advancements have seen these models adopt more complex, methodical reasoning strategies that parallel the dual-process theory of human cognition. Notably, approaches such as chain-of-thought (CoT) [155] and least-to-most prompting (L2M) [156] demonstrate this shift by encouraging a step-by-step reasoning that reflects deliberate human thought processes. AoT is the latest development in this evolution, integrating algorithmic reasoning into LLMs, enabling them to navigate complex problem spaces with a structured, strategic methodology that closely resembles human algorithmic thinking. At the heart of AoT is its ability to utilize in-context algorithmic examples, leveraging the generative capabilities of LLMs to explore various reasoning pathways and potential solutions efficiently. This method significantly cuts computational and environmental costs by reducing the reliance on extensive queries or external computational mechanisms. AoT not only follows but also enhances the algorithmic logic provided, indicating an innate ability in LLMs to blend their generative power with structured reasoning, leading to more effective and potentially innovative problem-solving approaches.

Despite these advancements, AoT faces challenges, particularly in balancing the depth of algorithmic exploration with the efficient use of tokens within LLMs. This balance is crucial to maintain the effectiveness of the AoT approach within the operational limits of generative capabilities in the model. The trajectory of AI-driven thought creation, especially through AoT, seems poised for further sophisticated developments. The aim is to create more nuanced, context-aware algorithms that enhance the problem-solving abilities of LLMs. Such progress could lead to models that not only more closely simulate human reasoning but also contribute unique, AI-driven insights to complex challenges. The ongoing development and refinement of AI-Driven Thought Creation, particularly via the AoT methodology, represent significant milestones in artificial intelligence, promising exciting new capabilities for LLMs in problem-solving and idea generation.

6. Conclusion

This survey provides a structured and comprehensive synthesis of Transformer-based architectures for time-series analysis, spanning theoretical foundations, model variants, and domain-specific applications. By organizing existing methods through a task-driven taxonomy and examining their temporal inductive biases, efficiency mechanisms, and scalability properties, we consolidate the diverse research landscape into a coherent reference framework. Our analysis highlights three key insights. First, attention-based models offer clear advantages in capturing multi-scale temporal dependencies, but their performance depends strongly on the choice of positional encoding, sparsity strategy, and sequence-level inductive bias. Second, computational overhead remains a central barrier in real-time and networked systems, motivating

architectures that combine linear-complexity attention, hierarchical encoders, or hybrid CNN–Transformer structures. Third, generalization across domains is still limited, especially under distribution shift, non-stationarity, and irregular sampling—conditions frequently encountered in operational networked environments. Building on these findings, we outline several promising research directions: lightweight and edge-deployable Transformer variants for latency-sensitive applications; cross-domain pretraining and foundation models for networked time series; robustness against concept drift and adversarial perturbations; and multimodal integration combining traffic, sensor, and system logs for richer representations. These directions align with the growing need for reliable, scalable, and real-time predictive intelligence in large networked and distributed systems. By consolidating the state of the art and identifying the technical gaps ahead, this survey aims to serve both researchers and practitioners as a foundation for advancing Transformer-based approaches in next-generation time-series modeling and networked applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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